BARREL: measure of volume equal to 42 US gallons; also name given to wrought-iron or wrought-steel gas-service pipes.

BATTER: sloping sides of a trench.

BED: ground or material on which a buried pipe is first laid.

BELL AND SPIGOT: corresponds to spigot and socket.

BELLOWS SEAL: corrugated seal form providing isolation of a valve stem.

BITUMEN/BITUMINOUS PRODUCTS: petroleum-based products used for coatings, linings, etc.

BLANK: solid plug, disc or end fitting for sealing an open-pipe end.

BLOW-DOWN: reducing pipeline (usually gas) pressure by venting to atmosphere.

BONDING: connecting of all metal parts in a system with a conductor.

BODY: the framework that holds together the parts of a valve.

BONNET: the case enclosing the stem on a screw-down valve.

BORE: internal diameter of a pipe or tube.

BRANCH: specifically refers to a connection to or from a main pipe to secondary pipes. Described under various types and construction, e.g. tee, Y, etc.; welded, forged, cast, etc.

BRANCH LINE: secondary pipeline(s) related to the main pipeline.

BULL PLUG: temporary closure or plug fitted to a pipeline under construction to prevent ingestion of dust, etc.

BURSTING DISC: pressure-relief device arranged to rupture and vent excess pressure to atmosphere.

BUTTERFLY VALVE: type of valve where the moving element is a disc mounted at right angles to the flow and pivoted in a place at right angles to the flow.

BUTTERFLY CHECK VALVE: similar geometry to a butterfly valve except that the disc is hinged about a diameter at right angles to the flow.

BYPASS: alternative flow passage to the main stream.

CAP: fitting which goes over the end of a pipe to seal it.

CAPPING VALVE: in pulp mills, a remotely operated shut-off valve used for chip feeding on batch digesters.

CARRIER PIPE: describes the pipe carrying the fluid in any installation where the pipe itself is surrounded by a sleeve or second pipe.

CATALYST: as distinct from a hardener, a catalyst is generally an organic peroxide which initiates polymerisation of polyester, vinyl ester resins.

CATHODIC PROTECTION: method of inhibiting corrosion by making system components in the system cathodic and confining corrosion to an attached sacrificial anode.

CHANGE FITTING: fitting for connecting Imperial (inch) size pipes to near-equivalent metric size pipes (i.e. nominally equivalent bore sizes).

CHECK VALVE: valves designed to shut off flow in one direction but permit free flow in the opposite direction. Also known as a non-return valve .

CLOSURE MEMBER: The moving component of a valve that throttles or shuts off flow through the valve body, such as a disc, a ball, etc.

COCK: general description of a small on–off valve, of which there are several basic types.

CODES OF PRACTICE: recommendations rather than obligatory requirements issued by national and international authorities.

COMPACTION: measure of the density of the soil at any given location.

CONDENSATE: liquid formed by wet air or gases, or vapours, when subject to cooling and/or pressure reduction.

CONTOUR LAYING: laying of underground pipelines at substantially constant buried depth, i.e. following the contours of the bend.

CONTROL VALVE: general description for a type of valve used for controlling flow or pressure and usually referred to by function, e.g. throttling valve, flow-control valve, pressure-control valve, etc.

COUPLING: fitting used to connect pipes.

COVER: the buried depth (i.e. depth below ground level) of a buried pipe or pipelines.

CRITICAL APPLICATIONS: applications or systems where failure of a pipe or valve could have serious consequences.

CRUDE LINE: pipeline for conveying crude oil.

CRYOGENIC SYSTEMS: systems whose components are designed to operate at and withstand extremely low temperatures. Generally descriptive of systems handling liquefied gases.

DEAD BAND: the range through which an input signal to a valve can be varied without initiating a response.

DEAD MAIN: a main pipeline not in use, e.g. not yet connected or temporarily or previously disconnected.

DIAPHRAGM VALVE: valve type in which the moving element is a flexible diaphragm.

DISC: refers to any disc-shaped element in a valve, as distinct from a plug shape, ball, poppet, etc.

DOG LEG: abrupt change in the direction of a pipeline.

DOWNSTREAM: any position in the direction of flow distant from the reference point involved.

EQUIVALENT LENGTH: friction or head loss generated by pipes, fittings, etc. expressed in terms of length of same diameter pipe having the same frictional losses.

ELBOW: a sharp-bend fitting with less radius than a normal pipe bend for the same degree of bend.

EXTENSION STEM: extended stem fitted to valves to facilitate operation under particular circumstances (e.g. on cryogenic valves to remove operating point from a low temperature region).

FALL: the gradient at which a pipeline is laid.

FEEDER: a main pipeline carrying fluid at a higher pressure than in the secondary distribution pipes.

FITTING: general description for couplings, etc., used on pipes and tubes. In some industries this may also include bends, valves, etc.

FLASHING: in liquid service, a phenomenon where the pressure of the medium falls below vapour pressure and does not recover above vapour pressure.

FLEXIBLE PIPE: any type of pipe which can flex or deform markedly without fracture.

FLOW CHARACTERISTIC: in control valves, the curve relating percentage of flow to percentage of valve travel.

FLOW CO-EFFICIENT: a constant, related to the geometry of a valve, for a given valve opening that is used to calculate the capacity of a control valve. Flow co-efficient C_v is defined using imperial /US measurement units, K_v using metric units.

FOOT VALVE: check valve fitted to the end of the suction pipe leading to a pump.

FULL FLOW: refers to a pipe flowing full of fluid, or more specifically to flow through a valve offering minimal restriction in the open position.

GATE VALVE: type of valve where the sealing element is in the form of a sliding plate, disc or 'gate'.

GEAR OPERATORS: type of mechanical actuator employed to assist in the opening and closing of large valves.

GENERAL PURPOSE VALVE: type of valve which is suitable for use in a variety of duties, e.g. shut-off, throttling, etc.

GLOBE VALVE- type of valve with a spherical or globe-shaped casing.

HAMMER-BLOW HANDWHEEL: handwheel with lugs fitted to large valves, enabling a hammer to be used to start valve opening, or effect tight closure.

HEADER: pipe, tank or fitting interconnecting a number of branch pipes.

HEAD: pressure exerted by a column of fluid expressed in tens of feet or metres of fluid height above a reference point.

HYDRANT: fitting or connection for attaching a hose to a water main (e.g. as in fire-fighting equipment).

HYDRAULIC OPERATOR: valve actuator operated by hydraulic pressure.

HYSTERESIS: the maximum difference in output value for any single input value during a calibration cycle, excluding errors due to dead band.

IMPRESSED CURRENT SYSTEM: a form of cathodic protection utilising the flow of electric current through a bonded metallic system.

INSIDE-SCREW VALVE: screw-down valve where the thread of the spindle is inside the bonnet.

ISOLATING JOINT: insulating joint between metallic pipes.

JOINT: general description for the various types of joints and couplings used to connect two pipes, or pipes to fittings, etc.

LIFT-CHECK VALVE: common type of check valve where the valve element lifts off its seat to open the valve.

LIMIT SWITCH: a device connected to an actuator or valve transmitting a signal when the valve reaches a pre-established position.

LINE: alternative description (used in the UK) to describe a pipeline, particularly in compressed air and hydraulic systems.

LNG: liquefied natural gas.

LPG: liquefied petroleum gas.

LUBRICATED VALVE: valve in which the moving element is lubricated by something other than the fluid being handled (e.g. lubricated plug valve).

MAIN: a principal or trunk supply pipeline.

MANIFOLD: a component designed to accept and provide interconnection between a number of pipes and/or valves, etc.

MARKER POST: post erected to show the position of a buried pipeline, cathodic protection test point, inspection point, etc.

MOLE PLOUGHING: method of making a hole to bury a small-diameter pipe using a tractor-mounted blade with a bullet-shaped foot.

NETWORK: complete system of transmission or distribution pipelines.

NOMINAL DIAMETER: approximate diameter size of pipes. May be reference to overall diameter or bore size.

NON-RETURN VALVE: see check valve.

OPERATOR: alternative name for a valve actuator.

OUTSIDE-SCREW VALVE: screw-down valve where the spindle thread is outside the bonnet and fully exposed.

ON STREAM: descriptive of a plant or system being operational in its normal way.

ORIFICE: a hole through which fluid can flow.

OVALITY: departure from a true circle in the actual cross-section of a pipe. Ovality is commonly defined as the difference between maximum and minimum diameter at a given section, divided by the mean diameter and expressed as a percentage.

PACKING: general description for the sealing material used in a gland, e.g. to seal valve stems.

PIG: piston-shaped device drawn through a pipeline to clean, gauge or inspect it. There are various types and shapes used for different purposes.

PINCH VALVE: type of valve embodying a flexible tube which is pinched together to close the valve.

PILOT VALVE: a small valve used to control supply to a larger valve to operate that valve.

PILOT-OPERATED VALVE: a larger valve operated indirectly from a pilot valve.

PIPE: general description for any reasonably long length of tubular form used to convey fluids.

PIPELINE: a continuous length of pipe or pipes forming a fluid transport system. The description may also include all ancillary equipment.

PLUG VALVE: type of valve where the movable sealing element is in the form of a plug, particularly descriptive of types of locks.

POSITIONER: a device for varying and keeping the actuator position in control-valve applications.

PRESSURE RECOVERY: The difference between the minimum pressure at the valve's vena contracta and the maximum pressure at the valve's outlet.

PRESSURE-REDUCING VALVE: a valve specifically designed to reduce steam, gas or liquid flow to a predetermined lower level.

PRESSURE-RELIEF DEVICE: A device designed to prevent internal fluid pressure from rising above a predetermined maximum pressure in a pressure vessel exposed to emergency or abnormal conditions.

PRESSURE-RELIEF VALVE: effectively a safety valve which can be used both with liquids (non-compressible fluids) and air, gas or vapours (compressible fluids).

PORT: opening in a valve through which fluid flows when the valve is open.

RUNNER: tool for consolidating backfill in trenches.

RELIEF VALVE: generally a slow-opening valve designed to relieve excess pressure in a liquid system.

REYNOLDS NUMBER: non-dimensional parameter which can be used to establish whether fluid flow is laminar or turbulent.

SAFETY VALVE: quick-opening valve providing pressure relief in system involving compressible fluids.

SEAT: the sealing face against which the moving or controlling element in a valve closes to provide shut-off.

SLIDE VALVE: valve whose ports are opened and closed by the sliding movement of a sleeve, disc, gate, etc.

SOCKET: the enlarged end of a pipe which fits over the plain end of a similar pipe, or a spigot (spigot and socket joint).

SOLENOID VALVE: a valve operated directly by an electromagnet or solenoid.

STACK PIPE: a vent.

STATIC PRESSURE RATING (pipe): normal operating conditions of pipe systems when connected to centrifugal or turbine pumps.

STANDPIPE: a vertical pipe used for withdrawing condensate (gas industry), or for providing a temporary supply of water at uniform pressure (water supply industry).

STEM: the spindle of a screw-down valve.

STOP COCK: description used specifically in the water industry for a shut-off valve.

STOPPING OFF: fitting of a temporary plug in a pipeline.

TAKE-OFF: a branch pipeline.

THERMOPLASTIC: a plastic material which can be transformed under the influence of heat and which solidifies upon cooling.

THERMOSET: a plastic which hardens when heated (and assisted by a chemical reaction) and which cannot be transformed or modified subsequent to this reaction.

THRUST BLOCK: an anchor block located against a bend or an end tap to prevent displacement of a pipe.

THROTTLING VALVE: a type of valve suitable for providing varying degrees of flow without creating excessive frictional losses.

TOP ENTRY: a construction of a valve body in which the valve is assembled and disassembled through the top of the valve.

TRIM: the replaceable parts of a valve.

TRUNNION MOUNTING: a construction used in valves where the trim is supported with bearings. The upstream spring-loaded seat normally acts as the primary seat.

TUBE: mainly descriptive of smaller diameter smooth-bore seamless pipe.

UNION: threaded fitting which can be used to connect two pipes without having to rotate either pipe.

UNION BONNET: type of bonnet used on smaller valves which can be assembled or disassembled by rotation of the union nut only.

VALVE: virtually any device with a mechanical movement used for controlling the flow or pressure of fluids.

VALVE ACTUATOR: a mechanism for operating valves which may be powered by compressed air, hydraulics or an electric motor.

VALVE RANGEABILITY: the ratio of the highest controllable flow coefficient to the lowest controllable flow coefficient (max C_v/min C_v).

VENA CONTRACTA: the location in a control valve where the cross-sectional area of the flowstream is at its smallest, fluid velocity at its highest and fluid pressure at its lowest level.

WALL THICKNESS: the thickness of the walls of a pipe or tube.

WHEEL OPERATOR: a handwheel attached to the top of a valve spindle for opening or closing the valve manually.

WIRE DRAWING: premature erosion of a valve seat caused by excessive flow velocities between the seat and the moving element of the valve.

WRAPPING: an outer layer of material applied to a pipe to protect the pipe surface.

Standards and Designations

Originators of American Standards

AAR (Association of American Railroads): establishes design and dimensional standards on bronze valves and 300 lb malleable pipe fittings for use by railroads.

API (American Petroleum Institute): establishes purchasing standards on valves and fittings for the petrochemical industry.

ASME (American Society of Mechanical Engineers): establishes codes covering pressure–temperature ratings, minimum wall thicknesses, metal specifications and performance, thread specifications, etc., for valves made of materials meeting ASTM specifications.

ASTM (American Society for Testing Materials): establishes chemical and physical requirements of all materials used in the manufacture of valves and fittings.

AWWA (American Water Works Association): established standards on iron gate valves to be used in a recognised water supply system.

Federal Specification (Federal Government Specification Standards): established by US agencies for design, dimensions, materials, and tests on items for use by the Armed Forces.

FM (AFM) (Associated Factory Mutual): establishes standards similar to UL but is employed by Mutual Fire Insurance Companies.

MIL Specifications (US Military Specifications and Standards): establishes design, dimensions, materials, and tests on items for use by the Armed Forces.

MSS (Manufacturers' Standardization Society of the Valves and Fittings Industry): maintains standards on dimensions, marking, boss locations for drains and bypasses, testing, and other similar type standards.

NFPA (National Fire Protection Association): establishes design and performance standards on valves and fittings used in fire protection service.

USASI (USAS) (United States of America Standards Institute): establishes certain basic dimensions of valves, fittings and threads.

UL (Underwriters Laboratories): establishes design and performance standards on valves and fittings used in fire protection service and handling of hazardous liquids.

ASTM metal and alloy designations

Bronzes and brasses

High tensile steam bronze	B-61
Steam bronze	B-62
Cast silicon (Everdur 1000)	B-198 grade 12A
Silicon brass	B-198 grade 13B
Silicon brass	B-371 alloy A
Wrought silicon (Everdur 1012)	B-98 alloy D
88-10-2 bronze	B-143 class 1A
Ampco, grade C-3	B-148 alloy 9C
Ampcoloy, grade B-2	B-148 alloy 9B
Brass rod	B-16
Naval brass	B-21 alloy A
Brass tubing	B-135 alloy G
Phosphor bronze	B-134 alloy B-2
Bronze rod	B-134 alloy B

Irons

Cast iron	A-126 class A
Cast iron	A-126 class B
High tensile cast iron	A-126 class C
Malleable cast iron	A-47 grade 35018
Malleable cast iron	A-47 grade 32510
Ni-resist grey cast iron	A-346 type II

Cast steels

Carbon steel, cast	A-126 grade WCB
0.15% Moly steel, cast	A-217 grade WC1
Cr Moly steel, cast	A-217 grade WC-6
Cr Moly steel, cast	A-217 grade WC-9
4.6% Cr Moly steel	A-217 grade C5
8–10% Cr Moly steel	A-217 grade C-12
Carbon steel, cast	A-325 grade LCB
Carbon Moly steel, cast	A-352 grade LC-1
3.5% Nickel steel, cast	A-352 grade LC-3

Stainless steels

18.8S cast	A-351 grade CF8 (type 304)
Wrought	A-276 type 304
Wrought	A-276 type 303
18-8S Mo, cast	A-351 grade CF-8M (type 316)
Wrought	A-276 type 316
18-8S Cb, cast	A-351 grade CF-8C (type 347)
Wrought	A-276 type 347
11.5–13.5 Cr steel	A-182 grade F-6
11.5–13.5 Cr steel	A-276 type 416
Heat-resisting 25-12, cast	A-297 grade MH
Heat-resisting 25-20, cast	A-297 grade MK
Heat-resisting 25-20, wrought	A-182 type F310
Heat-resisting 15-35	A-297 grade Ht

Nickel alloys

Nickel, cast	A-296 CZ100
Nickel, wrought	B-160
Monel, cast	A-296 M-35W
Monel, wrought	B-164 class A
Hastelloy 'B', cast	A-296 N-12M
Hastelloy 'C', wrought	A-296m CW-12M

Aluminium

No 356T, cast	B-26 grade SG70AT6

Standards, Specifications and Codes of Practice

Pipes and pipe fittings: British Standards

BS 21:1973 (1986): Pipe threads for tubes and fittings where pressure-tight joints are made on the threads (metric dimensions).
BS 65:1981: Vitrified clay pipes, fittings and joints.
BS 78: Cast-iron spigot and socket pipes (vertically cast) and spigot and socket fittings.
BS 78:Part 2:1965 (1981): Fittings.
BS 416:1973: Cast-iron spigot and socket soil, waste and ventilating pipes (sand-cast and spun) and fittings.
BS 437:1978: Specification for cast iron spigot and socket drain pipes and fittings.
BS 486: 1981: Asbestos–cement pressure pipes and joints.
BS 497: Part 1:1976: Cast iron and cast steel.
BS 534:1981: Steel pipes and specials for water and sewage.
BS 556:Part 1: 1966: Concrete cylindrical pipes and fittings, including manholes, inspection chambers and street gullies. Part 1: Imperial units.
BS 556:Part 2:1972: Concrete cylindrical pipes and fittings, including manholes, inspection chambers and street gullies. Part 2: Metric units.
BS 1194:1969: Concrete porous pipes for under-drainage.
BS 1196:1971 (1977): Clayware field-drain pipes.
BS 1211:1958 (1981): Centrifugally-cast (spun) iron pressure pipes for water, gas and sewage.
BS 1387:1967: Steel tubes and tubulars suitable for screwing to BS 21 pipe threads.
BS 1737:1951: Jointing materials and compounds for water, town gas and low-pressure steam installations.
BS 1965: Butt-welding pipe fittings for pressure purposes.
BS 1965:Part 1:1963 (1983): Carbon steel.
BS 1972:1967: Polythene pipe (type 32) for cold-water services.
BS 2494:1976: Materials for elastomeric joint rings for pipework and pipelines.
BS 2760:1973: Pitch-impregnated fibre pipes and fittings for below and above ground drainage.
BS 2815:1973: Compressed asbestos fibre jointing.
BS 2871:Part 1:1971: Copper tubes for water, gas and sanitation.
BS 3063:1965: Dimensions of gaskets for pipe flanges. (Note: This BS is used only in connection with obsolescent BS 10 flanges).
BS 3284:1967: Polythene pipe (type 50) for cold-water services.
BS 3505:1968 (1982): Unplasticised PVC pipe for cold-water services.
BS 3506:1969: Unplasticised PVC pipe for industrial purposes.
BS 3656:1981: Specification for asbestos–cement pipes, joints and fittings for sewage and drainage.

BS 4508: Thermally insulated underground piping systems.
BS 4508:Part 1:1969: Steel-cased systems with air gap.
BS 4622:1970: Grey iron pipes and fittings.
BS 4625:1970: Pre-stressed concrete pressure pipes (including fittings).
BS 4660:1973: Unplasticised PVC underground drain pipe and fittings.
BS 4772:1971: Ductile iron pipes and fittings.
BS 4962:1982: Specification for plastic pipes for use as light sub-soil drains.
BS 5178:1975: Pre-stressed concrete pipes for drainage and sewage.
BS 5911: Pre-cast concrete pipes and fittings for drainage and sewage.
BS 5911:Part 1:1981: Specification for concrete cylindrical pipes, bends, junctions and manholes, unreinforced or reinforced with steel cages or hoops.
BS 5911:Part 2:1982: Specification for inspection chambers and gullies.
BS 5911:Part 3:1982: Specification for ogee-jointed concrete pipes, bends and junctions, unreinforced or reinforced with steel cages or hoops.
BS 5955: Code of practice for plastic pipework (thermoplastic materials).
BS 5955:Part 6:1980: Installation of unplasticised PVC pipework for gravity drains and sewers.
BS 6087:1981: Specification for flexible joints for cast-iron drainpipes and fittings (BS 437) and for cast-iron soil, waste and ventilating pipes and fittings (BS 416).
BS 6464:1984: Specification for reinforced plastic pipes, fittings and joints for process plants.
CP 2010: Pipelines.
CP 2010:Part 1:1966: Installation of pipelines in land.
CP 2010:Part 2:1970: Design and construction of steel pipelines in land.
CP 2010:Part 3:1972: Design and construction of iron pipelines in land.
CP 2010:Part 4:1972: Design and construction of asbestos–cement pipelines in land.
CP 2010:Part 5: 1974: Design and construction of pre-stressed concrete pressure pipelines in land.
DD 76: Pre-cast concrete pipes of composite construction.
DD 76:Part 1:1981: Precast concrete pipes strengthened by continuous alkali-resistant glass rovings.

Valves and fittings: British Standards

BS 143 and 1256:1968: Malleable cast iron and cast copper alloy screwed pipe fittings for steam, air, water, gas and oil.
BS 1010: Draw-off taps and stop valves for water service (screwdown pattern).
BS 1010:Part 2:1973: Draw-off taps and above-ground stop valves
BS 1123:1976: Specification for safety valves, gauges and other safety fittings for air receivers and compressed-air installations.
BS 1414:1975 (1983): Steel wedge-gate valve (flanged and butt-welding ends) for the petroleum, petrochemical and allied industries.

BS 1553:Part 1:1977: Piping systems and plant.
BS 1640:1962 and 1968: Steel butt-welding pipe fittings for the petrochemical industry.
BS 1655:1950: Flanged automatic control valves for the process control industry (face-to-face dimensions).
BS 1868:1975 (1983): Steel check valves (flanged and butt-welding ends) for the petroleum, petrochemical and allied industries.
BS 1873:1975: Steel globe and globe stop and check valves (flanged and butt-welding ends) for the petroleum, petrochemical and allied industries.
BS 1963:1969: Pressure-operated relay valves for gas-burning appliances.
BS 2080:1974: Face-to-face, centre-to-centre, end-to-end and centre-to-end dimensions of flanged and butt-welding end steel valves for the petroleum, petrochemical and allied industries.
BS 2580:1979: Underground plug cocks for cold-water services.
BS 3016:1983: Pressure regulators and automatic changeover devices for liquefied petroleum gases.
BS 3059: Steel boiler and superheater tubes.
BS 3059:Part 1:1978 ±ISO 2604/II, ISO 2604/III, ISO 1129: Low tensile carbon steel tubes without specified elevated temperature properties.
BS 3059:Part 2:1978 ±ISO 2604/II, ISO 2604/3, ISO 1129: Carbon, alloy and austenitic stainless steel tubes with specified elevated temperature properties.
BS 3600:1976 ≠ISO 336, ±ISO 64: Dimensions and masses per unit length of welded and seamless steel pipes and tubes for pressure purposes.
BS 3604:1978 ±ISO 2604/II, ISO 2604/III, ISO 2605/I: Steel pipes and tubes for pressure purposes: ferritic alloy steel with specific elevated temperature properties.
BS 3799:1974: Steel pipe fittings, screwed and socket-welding for the petroleum industry.
BS 4504: Flanges and bolting for pipes, valves and fittings. Metric series.
BS 4504:Part 1:1969: Ferrous.
BS 4504:Part 2:1974: Copper alloy and composite flanges.
BS 4740: Method of evaluating control-valve capacity.
BS 4740:Part 1:1971: Incompressible fluids.
BS 5146:1974: Inspection and test of steel valves for the petroleum, petrochemical and allied industries.
BS 5150:1974: Cast-iron wedge and double-disc gate valves for general purposes.
BS 5151:1974 (1983): Cast-iron gate (parallel-slide) valves for general purposes.
BS 5152:1974 (1983): Cast-iron globe and globe stop and check valves for general purposes.
BS 5153:1974 (1983): Cast iron check valves for general purposes.
BS 5154:1983: Copper alloy globe, globe stop and check, check, and gate valves for general purposes.

BS 5155:1974: Cast-iron and carbon steel butterfly valves for general purposes.
BS 5156:1974 (1986): Screwdown diaphragm valves for general purposes.
BS 5157:1974: Steel gate (parallel slide) valves for general purposes.
BS 5158:1974: Cast- iron and carbon steel plug valves for general purposes.
BS 5159:1974: Cast-iron and carbon steel ball valves for general purposes.
BS 5160:1974: Flanged steel globe valves, globe stop and check valves.
BS 5163:1974:Double-flanged cast-iron wedge-gate valves for water-works purposes.
BS 5351:1986: Steel ball valves for the petroleum, petrochemical and allied industries.
BS 5352:1981 (1983): Steel wedge-gate, globe and check valves 50 mm and smaller for the petroleum, petrochemical and allied industries.
BS 5353:1980: Plug valves.
BS 5417:1976: Testing of general purpose industrial valves.
BS 5418:1979 = ISO 5209: Marking of general purpose industrial valves.
BS 5882:1980 ≠ISO/DIS 6215: A total quality-assurance programme for nuclear power plants.
BS 6683:1986: Guide to the installation and use of valves.
MA 65:Parts 1–11:1975–1977: General purpose and petroleum industry valves for use in marine pipeline systems.

SI units

SI units are the seven basic units and the units derived from them coherently, i.e. with numerical factor 1.

SI basic units

Basic quantity		SI basic unit	
Name	Symbol	Name	Symbol
Length	l	Metre	m
Mass	m	Kilogram	kg
Time	t	Second	s
Electric current	I	Ampere	A
Thermodynamic temperature	T	Kelvin	K
Amount of substance	n	Mole	mol
Luminous intensity	I_v	Candela	cd

Internationally defined prefixes

Meaning	Prefix: Name	Prefix: Symbol	Power of 10	Factor as decimal
Trillion	exa	E	10^{18}	= 1 000 000 000 000 000 000
Thousand billion	peta	P	10^{15}	= 1 000 000 000 000 000
Billion	tera	T	10^{12}	= 1 000 000 000 000
Thousand million	giga	G	10^{9}	= 1 000 000 000
Million	mega	M	10^{6}	= 1 000 000
Thousand	kilo	k	10^{3}	= 1 000
Hundred	hecto	h	10^{2}	= 100
Ten	deca	da	10^{1}	= 10
Tenth	deci	d	10^{-1}	= 0.1
Hundredth	centi	c	10^{-2}	= 0.01
Thousandth	milli	m	10^{-3}	= 0.001
Millionth	micro	µ	10^{-6}	= 0.000 001
Thousand millionth	nano	n	10^{-9}	= 0.000 000 001
Billionth	pico	p	10^{-12}	= 0.000 000 000 001
Thousand billionth	femto	f	10^{-15}	= 0.000 000 000 000 001
Trillionth	atto	a	10^{-18}	= 0.000 000 000 000 000 001

Units

Size	Symbol	SI units	Permissible units other than SI	Conversion into associated SI unit and ratios	No longer permissible units and conversions
Length	l	m (metre)			1″ (in = 0.254 m) 1) 1 nm (nautical mile) = 1852 m
Surface	A	m^2 (square metre)			1 b (barn = 10^{-28} m^2 2) 1 a (are) = 10^2 m^2 2) 1 ha (hectare) = 10^4 m^2 sq.m., sq.dm., sq.cm., etc. – Name allowed, symbol not allowed
Volume	V	m^3 (cubic metre)	l (litre)	$1\,l = 10^{-3}\,m^3$	

Value	Symbol	SI units	Permissible units other than SI	Conversion into relevant SI unit and ratios	No longer permissible units with conversions
Solid angle	Ω	sr (steradian)		1 sr = $1 m^2/m^2$	1 □° (square degree) = $3.046 \cdot 10^{-4}$ sr 1 □g (square grade) = $2.467 \cdot 10^{-4}$ sr
Time	t	s (second)	min (minute) h (hour) d (day)	1 min = 60 s 1 h = 3600 s 1 d = 86.400 s	
Frequency	f	Hz (Hertz)		1 Hz = 1/s	
Rotation speed	n	s^{-1}	rpm rpm	1 rpm = $(^1/_{60})\, s^{-1}$ 1 rpm = 1 (1/min)	
Velocity	v	m/s	km/h	1 km/h = (1/3.6) m/s	
Acceleration	@	m/s^2		Normal fall acceleration g_n=9.80665 m/s^2	1 gal (gal) = 10^{-2} m/s^2
Mass	m	kg (kilogram)	t (tonne)	1 t = 10^3 kg	1 q (metric hundredweight) = 100 kg
Density	p	kg/m^3	t/m^3 kg/l	1 t/m^3 = 1000 kg/m^3 1 kg/l = 1000 kg/m^3	
Moment of inertia	J	$kg \cdot m^2$			1 $kp \cdot m \cdot s^2$ = 9.81 $kg \cdot m^2$
Force	F	N (Newton)		1 N=1 $kg \cdot m/s^2$	1 dyn (dyn) = 10^{-5} N 1 p (pond) = $9.80665 \cdot 10^{-3}$ N 1 kp (kilopond) = 9.80665 N
Torque	M	N·m			1 kpm=9.80665 Nm
Pressure	p	Pa (Pascal)	bar	1 Pa = 1 N/m^2 1 bar = 10^5 Pa	1 atm = 1.01325 bar 1 at = 0.980665 bar 1 Torr = $1.333224 \cdot 10^{-3}$ bar 1 m CE = $98.0665 \cdot 10^{-3}$ bar 1 mm Hg = $1.333224 \cdot 10^{-3}$ bar
Mechanical stress	R	N/m^2 Pa		1 N/m^2 = 1 Pa	1 kp/m^2 = 9.80665 N/m^2 1 kp/cm^2 = $98.0665 \cdot 10^{-3}$ N/m^2 1 kp/mm^2 = $98.0665 \cdot 10^{-6}$ N/m^2
Dynamic viscosity	✂	Pa·s		1 Pa·s = 1 $N \cdot s/m^2$	1 P (poise) = 10^{-1} Pa·s
Kinematic viscosity	⊗	m^2/s		1 m^2/s = 1 $Pa \cdot s \cdot m^3/kg$	1 St (stokes) = 10^{-4} m^2/s
Work Energy	W E	J (Joule)	eV (electronvolt) W·h	1 J = 1 Nm = 1 H_2O 1 W·h = 3.6 KJ	1 cal = 4.1868 J 1 kpm = 9.80665 J 1 erg = 10^{-4} J
Electric charge	Q	C (Coulomb)		1 C = 1 A·s	
Electric potential	U	V (Volt)		1 V = 1 W/A	
Electric current	I	A (Ampere)			
Electric resistance	R	Ω (Ohm)		1 Ω = 1 V/A	1 Ω abs = 1 Ω
Power	P	W (Watt)		1 W = 1 J/s = 1 Nm/s = 1 V·A	1 PS = 735.498 W 1 kcal/h = 1.163 W 1 kpm/s = 10 W
Electric capacitance	C	F (Farad)		1 F = 1 C/V	
Magnetic field strength	H	A/m			1 Oe (oersted) = 79.5775 A/m
Magnetic flux	ϕ	Wb (Weber)		1 Wb = 1 V·s	1 Mx (maxwell) = 10^{-8} Wb
Magnetic flux density	B	T (Tesla)		1 T = 1 Wb/m^2	1 G (gauss) = 10^{-4} T
Inductance	L	H (Henry)		1 H = 1 Wb/A	
Electric conductance	G	S (Siemens)		1 S = 1/Ω	
Spec. electric resistance	σ	Ω/m			
Thermodynamic temperature	T	K (Kelvin)		Δ 1°C = 1 K 0°C ≙ 273.15 K	
Temperature (°C)	t/δ	°C (degree Celsius)		Δ 1°C = 1 K 0 K ≙ −273.15 °C	
Thermal capacity	C	J/K			1 Kcl/degree = $4.1868 \cdot 10^{-3}$ J/K 1 Cl (clausius) = 4.1868 J/K

Ball valves

Typical standards of compliance

	Specification	Description
Design	BS 5351	Specification for steel ball valves for the petroleum, petrochemical and allied industries
	BS 1560	Class designated flanges
	BS 4504	Valves: flanged ends
	ANSI B16.5	Valves: flanged ends
	ANSI B16.24	Valves: flanged ends
	ISO 5211	Valve/actuator mating dimensions
	API 6D	Specification for pipeline valves (gate, plug, ball and check valves)
Test	BS 6755 Part 1	Specification for production pressure-testing requirements
	BS 6755 Part 2	Specification for fire type-testing requirements
	API 6FA	Specification for fire type-testing requirements
	API 6D	Specification for pipeline valves
	API 598	Valve inspection and test
	API 607	Fire test
	ISO 5208	Industrial valve pressure testing
	NES 729	Requirements for non-destructive examination methods. Part 1: radiographic
	ASME V SE 165	Requirements for non-destructive examination methods. Dye penetrant

ASTM test methods

C 177-85	Test method for steady-state heat flux measurements and thermal transmisssion properties by means of the guarded-hot-plate apparatus: 04.06,08.01,14.01.
D 149-81	Test methods for dielectric breakdown voltage and dielectric strength of solid electrical insulating materials at commercial power frequencies: 08.01,09.02,10.02.
D 150-81	Test methods for A-C loss characteristics and permittivity (dielectric constant) of solid electrical insulating materials: 08.01,09.02,10.02,10.03.
D 256-84	Test methods for impact resistance of plastics and electrical insulating materials: 08.01,09.02.
D 570-81	Test method for water absorption of plastics: 08.01.
D 635-81	Test method for rate of burning and/or extent and time of burning of self-supporting plastics in a horizontal position: 08.01.
D 638-84	Test method for tensile properties of plastics: 08.01.
D 648-82	Test method for deflection temperature of plastics under flexural load: 08.01.
D 695-85	Test method for compressive properties of rigid plastics: 08.01.
D 696-79	Test method for coefficient of linear thermal expansion of plastics: 08.01,14.01.
D 790-84a	Test methods for flexural properties of unreinforced and reinforced plastics and electrical insulating materials: 08.01.
D 791	Discontinued: replaced by E 308.
D 792-66 (1979)	Test methods for specific gravity and density of plastics by displacement: 08.01.
D 1784-81	Specification for rigid poly(vinyl chloride) (PVC) compounds and chlorinated poly(vinyl chloride) (CPVC) compounds: 08.02,08.04.
D 2240-86	Test method for rubber property: durometer hardness: 08.02,09.01.
D 2766-83	Test method for specific heat of liquids and solids: 05.02.
D 3915-80	Specification for poly(vinyl chloride) (PVC) and related plastics pipe and fitting compounds: 08.03.08.04.
E 84-84	Test method for surface burning characteristics of building materials: 04.07.
E 162-83	Test method for surface flammability of materials using a radiant heat energy source: 04.07.
E 308-85	Method for computing the colours of objects by using the CIE system: 14.02.

List of relevant standards

Polyethylene piping systems

ISO R161	Thermoplastic pipes for the transport of fluids.
ISO 1183	Polyethylene: measurement of density.
ISO 3607	PE pipes: tolerances and o.d. and wall thicknesses.
ISO 3663	PE pressure pipes and fittings: dimensions of flanges.
ISO 4437	Buried PE pipes for the supply of gaseous fuels.
ISO 4440	PE pipes and fittings: determination of melt flow rate.
ISO 6447	Rubber seals: joint rings used for gas supply pipes and fittings.
DIN 3535	Seals for gas supply.
DIN 3543	PE-HD valves for pipes made from PE-HD material: dimensions.
DIN 3544	Valves in high-density polyethylene (PE-HD): specification and testing of tapping valves.
DIN 8074	Pipes in high-density polyethylene (PE-HD): dimensions.
DIN 8075	Pipes in high-density polyethylene (PE-HD). General quality requirements: testing.
DIN 16963	Pipe joints and piping components for pressure pipelines in high-density polyethylene (PE-HD).
DIN 19533	Pipes in PE-HD and PE-MD for drinking water supply: pipes, pipe joints, piping components.
DS 2131.2	Pipes, fittings and joints of PE-type PEM and PEH for buried gas pipelines.
DVS 2207 Part 1	Welding of thermoplastic materials, (PE) pipes and pipeline components for gas and water pipelines.
DVGW G 477	Manufacture, quality assurance and testing of pipes in rigid PVC and PE-HD for gas pipelines.
DVGW VP304	Gas tapping valves for PE-HD piping systems.
DVGW VP607	PE-HD fittings for gas and drinking water pipelines.
DVGW VP608	Polyethylene pipes (PE80 and PE100) for gas and drinking water lines: requirements and tests.
Ö Norm B 5192	Pipes, pipe joints and piping components in PE for buried gas pipelines.
prEN 1555	Plastic piping systems for gas distribution: polyethylene (PE).
prEN 12201	Plastic piping systems for water distribution: polyethylene (PE).
UNI 8849	Raccordi di polietilene (PE50), saldabili per fusione mediante elementi riscaldanti, per condotte per convogliamento di gas combustibili. Tipi, dimensioni e requisiti.
UNI 8850	Raccordi di polietilene (PE50), saldabili per elettrofusione per condotte interrate per convogliamento di gas combustibili: tipi, dimensioni e requisiti.

PVC piping systems

ISO 2045	Minimum insertion depth for push-fit sockets.
ISO 3460	PVC adaptor for backing flange.
ISO 3603	Leak test under internal pressure.
ISO/DIN 4422	PVC pipes and fittings for water supply.

DIN 2501 Part 1	Flange: connecting dimensions.
DIN 3441 Part 1	PVC valves: requirements and testing.
DIN 3543 Part 3	PVC tapping valves: dimensions.
DIN 4279 Part 7	Internal pressure testing of PVC pressure pipelines for water.
DIN 8061 Part1	PVC pipes: general quality requirements.
DIN 8062	PVC pipes: dimensions.
DIN 8063 Part 4	Pipe joints and components for PVC pressure pipelines: adaptors, flanges, gaskets, dimensions.
DIN 8063 Part 5	Pipe joints and components for PVC pressure pipelines: general quality requirements and test methods.
DIN 16450	Fittings for PVC pressure pipelines: designations, symbols.
DIN 16929	Chemical resistance of PVC.
DIN 19532	PVC pipelines for drinking water supply.
KRV A 1.1.2	Push-fit joints on PVC pressure pipes and fittings: dimensions, requirements, testing.
KIWA BRL K 603	Plastic gate valves of nominal sizes from 25 to 150 mm.
Quality specification No. 53	Couplings and fittings of unplasticised polyvinylchloride for water pipes.
Criteria No. 23	Doop Spuitgieten vervaardigde PVC-hulpstukken met flensaansluitigen.
BRL 2013	Rubberringen and flenspakkringen voor verbindungen in drinkwater en afvalwaterleidingen.
WIS 4-31-07	Specification for emplasticised PVC pressure fittings and assemblies for cold potable water (underground use).

Wear- and galling-resistance chart of material combinations

	304 SST	316 SST	Bronze	Inconel	Monel	Hastelloy B	Hastelloy C	Titanium 75A	Nickel	Alloy 20	Type 416 hard	Type 440 hard	17-4 PH	Alloy 6 (Co–Cr)	ENC*	Cr plate	Al bronze
304 SST	P	P	F	P	P	P	F	P	P	P	F	F	F	F	F	F	F
316 SST	P	P	F	P	P	P	F	P	P	P	F	F	F	F	F	F	F
Bronze	F	F	S	S	S	S	S	S	S	S	F	F	F	F	F	F	F
Inconel	P	P	S	P	P	P	F	P	F	F	F	F	F	F	F	F	S
Monel	P	P	S	P	P	P	F	F	F	F	F	F	F	S	F	F	S
Hastelloy B	P	P	S	P	P	P	F	F	S	F	F	F	F	S	F	S	S
Hastelloy C	F	F	S	F	F	F	F	F	F	F	F	F	F	S	F	S	S
Titanium 75A	P	P	S	P	F	F	F	P	F	F	F	F	F	S	F	F	S
Nickel	P	P	S	F	F	S	F	F	P	P	F	F	F	S	F	F	S
Alloy 20	P	P	S	F	F	F	F	F	P	P	F	F	F	S	F	F	S
Type 416 hard	F	F	F	F	F	F	F	F	F	F	F	F	F	S	S	S	S
Type 440 hard	F	F	F	F	F	F	F	F	F	F	S	F	S	S	S	S	S
17-4 PH	F	F	F	F	F	F	F	F	F	F	F	S	P	S	S	S	S
Alloy 6 (Co–Cr)	F	F	F	F	S	S	S	S	S	S	S	S	S	F	S	S	S
ENC*	F	F	F	F	F	F	F	F	F	F	S	S	S	S	P	S	S
Cr plate	F	F	F	F	F	S	S	S	S	S	S	S	S	S	S	P	S
Al bronze	F	F	F	S	S	S	S	S	S	S	S	S	S	S	S	S	P

*Electroless nickel coating.

S, satisfactory; F, fair; P, poor.

Valve-trim material temperature limits

	Lower		Upper	
Material	°C	°F	°C	°F
Type 304 stainless steel	−268	−450	316	600
Type 316 stainless steel	−268	−450	316	600
Bronze	−273	−460	232	450
Inconel[1]	−240	−400	649	1200
K Monel[1]	−240	−400	482	900
Monel	−240	−400	482	900
Hastelloy B[2]	—	—	371	700
Hastelloy C[2]	—	—	538	1000
Titanium	—	—	316	600
Nickel	−198	−325	316	600
Alloy 20	−46	−50	316	600
Type 416 stainless steel 40RC	−29	−20	427	800
CA-6NM	−29	−20	427	800
Nitronic 50[3]	−198	−325	538	1000
Type 440 stainless steel 60RC	−29	−20	427	800
17-4 PH (CB-7CU)	−40	−40	427	800
Alloy 6 (Co–Cr)	−273	−460	816	1500
Electroless nickel plating	−268	−450	427	800
Chrome plating	−268	−450	593	1100
Aluminium bronze	−273	−460	316	600
Nitrile	−40	−40	93	200
Fluoroelastomer (Viton[4] and Fluorel[5])	−23	−10	204	400
TFE	−268	−450	232	450
Nylon	−73	−100	93	200
Polyethylene	−73	−100	93	200
Neoprene	−40	−40	82	180

[1]Trademark of International Nickel Company.
[2]Trademark of Stellite Division, Cabot Corporation.
[3]Trademark of Armco Steel Corporation.
[4]Trademark of E.I. Du Pont de Nemours & Company, Inc.
[5]Trademark of 3M Company.

US standard materials for trim parts of valves

Material	Specification	Minimum physical properties				Modulus of elasticity at 70°F ($lb/in^2 \times 10^6$)	Hardness (Brinell)
		Tensile (lb/in^2)	Yield (lb/in^2)	Elongation in 2 in (%)	Reduction of area (%)		
Aluminium bar	ASTM B211 alloy 2011-T3	44,000	36,000	15	—	10.2	95
Yellow brass bar	ASTM B16 $^1/_2$ hard	45,000	15,000	7	50	14	—
Naval brass bar	ASTM B21 alloy 464	60,000	27,000	22	55	—	—
Leaded steel bar	AISI 12L14	79,000	71,000	16	52	—	163
Carbon steel bar	ASTM A108 grade 1018	69,000	48,000	38	62	—	143
Chrome-moly steel	AISI 4140 (suitable for ASTM A193 grade B7 bolt mat)	135,000	115,000	22	63	29.9	255
Type 302 stainless steel	ASTM A276 type 302	85,000	35,000	60	70	28	150
Type 304 stainless steel	ASTM A276 type 304	85,000	35,000	60	70	—	149
Type 316 stainless steel	ASTM A276 type 316	80,000	30,000	60	70	28	149
Type 316L stainless steel	ASTM A276 type 316L	81,000	34,000	55	—	—	146
Type 410 stainless steel	ASTM A276 type 410	75,000	40,000	35	70	29	155
Type 17-4PH stainless steel	ASTM A416 grade 630	135,000	105,000	16	50	29	275–345
Nickel–copper alloy bar	Alloy K500 (K Monel)	100,000	70,000	35	—	26	175–260
Nickel–moly alloy 'B' bar	ASTM B335 (Hastelloy 'B')	100,000	46,000	30	—	—	—
Nickel–moly–chrome alloy 'C' bar	ASTM B336 (Hastelloy 'C')	100,000	46,000	—	—	—	—

US standard materials for valve bodies (pressure-containing castings)

Material	Specification	Temperature range (°F)	Minimum physical properties				Modulus of elasticity at 70°F ($lb/in^2 \times 10^6$)	Hardness (Brinell)
			Tensile (lb/in^2)	Yield (lb/in^2)	Elongation in 2 in (%)	Reduction of area (%)		
Carbon steel	ASTM A216 grade WCB	−20 to 1000	70,000	36,000	22	35	27.9	137–187
Carbon steel	ASTM A352 grade LCB	−50 to 650	65,000	35,000	24	35	27.9	137–187
Chrome moly steel	ASTM A217 grade C5	−20 to 1100	90,000	60,000	18	35	27.4	241 max
Carbon moly steel	ASTM A217 grade WC1	−20 to 850	65,000	35,000	24	35	29.9	215 max
Chrome moly steel	ASTM 217 grade WC6	−20 to 1000	70,000	40,000	20	35	29.9	215 max
Chrome moly steel	ASTM A217 grade WC9	−20 to 1050	70,000	40,000	20	35	29.9	241 max
$3^1/_2$% nickel steel	ASTM A352 grade LC3	−150 to 650	65,000	40,000	24	35	27.9	137
Chrome moly steel	ASTM A217 grade C12	−20 to 1100	90,000	60,000	18	35	27.4	180–240
Type 304 stainless steel	ASTM A351 grade CF-8	−425 to 1500	65,000	28,000	35	—	28.0	140
Type 316 stainless steel	ASTM A351 grade CF-8M	−425 to 1500	70,000	30,000	30	—	28.3	156–170

Cast iron	ASTM A126 class B	−150 to 450	31,000	—	—	—	—	160–220
Cast iron	ASTM A126 class C	−150 to 450	41,000	—	—	—	—	160–220
Ductile iron	ASTM A395 type 60-45-15	−20 to 650	60,000	45,000	15	—	23–26	143–207
Ductile Ni-resist* iron	ASTM A439 type 60-45-15	−20 to 750	58,000	30,000	7	—	—	148–211
Standard valve bronze	ASTM B62	−325 to 450	30,000	14,000	20	17	13.5	55–65*
Tin bronze	ASTM B143 alloy 1A	−325 to 400	40,000	18,000	20	20	15	75–85*
Manganese bronze	ASTM B147 alloy BA	−325 to 350	65,000	25,000	20	20	15.4	98*
Aluminium bronze	ASTM B148 alloy 9C	−325 to 500	75,000	30,000	12 min	12	17	150
Monel† alloy 411	(Weldable grade)	−325 to 900	65,000	32,500	25	—	23	120–170
Nickel-moly alloy 'B'	ASTM A494 (Hastelloy 'B')	−325 to 700	72,000	46,000	6	—	—	—
Nickel-moly-chrome alloy 'C'	ASTM A494 (Hastelloy 'C')	−325 to 1000	72,000	46,000	4	—	—	—
Cobalt-base alloy No. 6	Stellite no. 6	—	121,000	64,000	1 to 2	—	30.4	—

*500 kg load;
†for Brinell and material.

Conversion tables

Viscosities

Kinematic viscosity centistokes density 1.0	Absolute viscosity centipoise	Engler °	Saybolt universal sec	Redwood 1 sec (standard)	Saybolt sec fural	Ford cup no. 4 furol	Barbey °	Cup no. 15 sec	Absolute viscosity poise density 1.0	Kinematic viscosity m^2/s
1.0	1.0	1.0	31	29	—	—	—	—	0.01	1.0×10^{-6}
2.0	2.0	1.1	34	30	—	—	3640	—	0.02	2.0×10^{-6}
3.0	3.0	1.2	35	33	—	—	2426	—	0.03	3.0×10^{-6}
4.0	4.0	1.3	37	35	—	—	1820	—	0.04	4.0×10^{-6}
5.0	5.0	1.39	42	38	—	—	1300	—	0.05	5.0×10^{-6}
6.0	6.0	1.48	45.5	40.5	—	—	1085	—	0.06	6.0×10^{-6}
7.0	7.0	1.57	48.5	43	—	—	930	—	0.07	7.0×10^{-6}
8.0	8.0	1.65	53	46	—	—	814	—	0.08	8.0×10^{-6}
9.0	9.0	1.74	55	48.5	—	—	723	—	0.09	9.0×10^{-6}
10	10	1.84	59	52	—	—	650	—	0.10	1.0×10^{-5}
20	20	2.9	97	85	15	—	320	—	0.2	2.0×10^{-5}
40	40	5.3	185	163	21	—	159	—	0.4	4.0×10^{-5}
60	60	7.9	280	245	30	18.7	106	5.6	0.6	6.0×10^{-5}
80	80	10.5	370	322	38	25.9	79	6.7	0.8	8.0×10^{-5}
100	100	13.2	472	408	47	32	65	7.4	1.0	1.0×10^{-4}
200	200	26.4	944	816	92	60	32.5	11.2	2.0	2.0×10^{-4}
400	400	52.8	1888	1632	184	111	15.9	18.4	4.0	4.0×10^{-4}
600	600	79.2	2832	2448	276	162	10.6	26.9	6.0	6.0×10^{-4}
800	800	106	3776	3264	368	217	8.1	35	8.0	8.0×10^{-4}
1000	1000	132	7080	4080	460	415	6.6	68	10	1.0×10^{-3}
5000	5000	660	23,600	20,400	2300	1356	1.23	240	50	5.0×10^{-3}
10,000	10,000	1320	47,200	40,800	4600	2713	—	481	100	1.0×10^{-2}
50,000	50,000	6600	236,000	204,000	23,000	13,560	—	2403	500	5.0×10^{-2}

Absolute viscosity (centipoise) = kinematic viscosity (centistokes) × density.
Over 50 centistokes, conversion to SSU: SSU = centistokes × 4.62.

Delivery volumes

m^3/h	l/min	hl/h	Imperial gallon/min	US gallon/min	cu ft/hr	cu ft/sec	m^3/sec
1.0	16.667	10.0	3.6667	4.3999	35.315	9.81×10^{-3}	2.78×10^{-4}
0.060	1.0	0.60	0.22	0.2642	2.1189	5.88×10^{-4}	1.67×10^{-5}
0.10	1.6667	1.0	0.3667	0.4399	3.5315	9.81×10^{-4}	2.78×10^{-5}
0.2727	4.546	2.7270	1.0	1.201	9.6326	1.67×10^{-3}	7.57×10^{-5}
0.2273	3.785	2.2732	0.8326	1.0	8.0208	2.23×10^{-3}	6.31×10^{-5}
0.0283	0.4719	0.2832	0.1038	0.1247	1.0	2.78×10^{-4}	7.86×10^{-5}
101.94	1699	1019.4	373.73	448.83	3600	1.0	0.0282
3600.0	6×10^{-4}	36,000	1320	15,838	127,208	35.315	1.0

hl = hectolitre = litre × 10^2.

Pressure and pressure heads

bar	kg/cm^2	lbf/in^2	atm	ft H_2O	m H_2O	mm Hg	in Hg	kPa
1.0	1.0197	14.504	0.9869	33.445	10.197	750.06	29.530	100
0.9807	1.0	14.223	0.9878	32.808	10	735.56	28.959	98.0
0.0689	0.0703	1.0	0.0609	2.3067	0.7031	51.715	2.036	6.89
1.0133	1.0332	14.696	1.0	33.889	10.332	760.0	29.921	101.3
0.0299	0.0305	0.4335	0.0295	1.0	0.3048	22.420	0.8827	2.99
0.0981	0.10	1.422	0.0968	3.2808	1.0	73.356	2.896	9.81
13.3×10^{-4}	0.0014	0.0193	13.2×10^{-4}	0.0446	0.0136	1.0	0.0394	0.133
0.0339	0.0345	0.4912	0.0334	1.1329	0.3453	25.40	1.0	3.39
1.0×10^{-5}	10.2×10^{-6}	14.5×10^{-5}	9.87×10^{-6}	3.34×10^{-4}	10.2×10^{-5}	75.0×10^{-4}	29.5×10^{-5}	1.0

atm = international standard atmosphere.
kg/cm^2 = metric atmosphere.

Velocity

metre per second m/s	foot per second ft/s	foot per minute ft/m	kilometre per hour km/hr	mile per hour mile/hr
1	3.2808	0.0547	3.6	2.2369
0.3048	1	0.0167	1.0973	0.6818
18.288	60	1	65.8368	40.9091
0.2778	0.9113	0.0152	1	0.6214
0.4470	1.4667	0.0245	1.6093	1

Volume

cubic metre m^3	cubic centimetre cm^3	litre l	cubic inch in^3	cubic foot ft^3	UK gallon UK gal	US gallon US gal
1	1,000,000	999.972	61,023.7	35.3147	219.969	264.172
0.000001	1	0.0009997	0.0610	0.0000353	0.00022	0.00026
0.001	1000.028	1	61.0255	0.0353	0.22	0.2642
0.000016	16.3871	0.0164	1	0.00058	0.0036	0.0043
0.0283	28316.8	28.3161	1728	1	6.2288	7.4805
0.0045	4546.09	4.546	277.419	0.1605	1	1.201
0.0038	3785.41	3.7853	231	0.1337	0.8327	1

Mass

kilogram kg	pound lb	hundredweight cwt	tonne t	UK ton	US short ton sh ton
1	2.2046	0.0197	0.001	0.00098	0.0011
0.4536	1	0.0089	0.000454	0.000446	0.0005
50.8023	112	1	0.0508	0.05	0.056
1000	2204.62	19.6841	1	0.9842	1.1023
1016.05	2240	20	1.0161	1	1.12
907.185	2000	17.8571	0.9072	0.8929	1

Density

gram per millilitre g/ml	kilogram per cubic metre kg/cm^3	pound per cubic foot lb/ft^3	pound per cubic inch lb/in^3
1	1000	62.428	0.0361
0.001	1	0.062	0.000036
0.016	16.02	1	0.00058
27.6807	27679.9	1728	1

Heat flow rate

watts W	calorie per second cal/s	kilocalorie per hour kcal/hr	British thermal unit per hour Btu/hr
1	0.2388	0.8598	3.4121
4.1868	1	3.6	14.286
1.163	0.2778	1	3.9683
0.2931	0.07	0.252	1

Force

kilonewton kN	kilogram force kgf	pound force lbf	poundal pdl
1	101.972	224.809	7233.01
0.00981	1	2.2046	70.9316
0.0044	0.4536	1	32.1740
0.000138	0.0141	0.8311	1

Torque

newton metre Nm	kilogram force metre kgf m	pound force foot lbf ft	pound force inch lbf/in
1	0.102	0.7376	8.8508
9.8067	1	7.2330	86.7962
1.3558	0.1383	1	12
0.113	0.0115	0.0833	1

Power

watt W	kilogram force metre per second kgf m/sec	metric horse power	foot pound force per second ft lbf/sec	horse power hp
1	0.102	0.00136	0.7376	0.00134
9.8067	1	0.01333	7.2330	0.01315
735.499	75	1	542.476	0.98632
1.3558	0.1383	0.00184	1	0.00182
745.70	76.0402	0.0139	550.0	1

Mass volumetric rate of flow: liquids

$$\text{US gal/min} = \frac{\text{lb/hr}}{500 \times SG_1} \qquad \text{m3/h} = \frac{0.001\text{kg/h}}{SG_2}$$

$$SG_1 = \text{water} = 1 \text{ at } 60^\circ\text{F} \qquad SG_2 = \text{water} = 1 \text{ at } 4^\circ\text{Celcius}$$

Linear conversions

Fractions to decimals to millimetres

Inch	Decimal-inch	Millimetre	Inch	Decimal-inch	Millimetre
	0.003937	0.1	15/32	0.46875	11.9063
	0.007874	0.2		0.472441	12
	0.011811	0.3	31/64	0.484375	12.3031
1/64	0.015625	0.3969	1/2	0.500	12.700
	0.015748	0.4		0.511811	13
	0.019685	0.5	33/64	0.515625	13.0969
	0.023622	0.6	17/32	0.53125	13.4938
	0.027559	0.7	35/64	0.546875	13.8906
1/32	0.03125	0.7938		0.551181	14
	0.031496	0.8	9/16	0.5625	14.2875
	0.035433	0.9	37/64	0.578125	14.6844
	0.03937	1		0.590551	15
3/64	0.046875	1.1906	19/32	0.59375	15.0813
1/16	0.0625	1.5875	39/64	0.609375	15.4781
5/64	0.078125	1.9844	5/8	0.625	15.875
	0.078740	2		0.629921	16
3/32	0.09375	2.3813	41/64	0.640625	16.2719
7/64	0.109375	2.7781	21/32	0.65625	16.6688
	0.118110	3		0.669291	17
1/8	0.125	3.175	43/64	0.671875	17.0656
9/64	0.140625	3.5719	11/16	0.6875	17.4625
5/32	0.15625	3.9688	45/64	0.703125	17.8594
	0.157480	4		0.708661	18
11/64	0.171875	4.3856	23/32	0.71875	18.2563
3/16	0.1875	4.7625	47/64	0.734375	18.6531
	0.196850	5		0.748031	19
13/64	0.203125	5.1594	3/4	0.750	19.050
7/32	0.21875	5.5563	49/64	0.765625	19.4469
15/64	0.234375	5.9531	25/32	0.78125	19.8438
	0.236220	6		0.787402	20
1/4	0.250	6.350	51/64	0.796875	20.2406
17/64	0.265625	6.7469	13/16	0.8125	20.6375
	0.275591	7		0.826772	21
9/32	0.28125	7.1438	53/64	0.828125	21.0344
19/64	0.296875	7.5406	27/32	0.84375	21.4313
5/16	0.3125	7.9375	55/64	0.859375	21.8281
	0.314961	8		0.866142	22
21/64	0.328125	8.3344	7/8	0.875	22.225
11/32	0.34375	8.7313	57/64	0.890625	22.6219
	0.354331	9		0.905512	23
23/64	0.359375	9.1281	29/32	0.90625	23.0188
3/8	0.375	9.525	59/64	0.921875	23.4156
25/64	0.390625	9.9219	15/16	0.9375	23.8125
	0.393701	10		0.944882	24
13/32	0.40625	10.3188	61/64	0.953125	24.2094
27/64	0.421875	10.7156	31/32	0.96875	24.6063
	0.433071	11		0.984252	25
7/16	0.4375	11.1125	63/64	0.984375	25.0031
29/64	0.453125	11.5094	1 in	1	25.400

Millimetres to inches

Milli-metres	Inches									
	0	1	2	3	4	5	6	7	8	9
0	—	0.03937	0.07874	0.11811	0.15748	0.19685	0.23622	0.27559	0.31496	0.35433
10	0.39370	0.43307	0.47244	0.51181	0.55118	0.59055	0.62992	0.66929	0.70866	0.74803
20	0.78740	0.82677	0.86614	0.90551	0.94488	0.98425	1.02362	1.06299	1.10236	1.14173
30	1.18110	1.22047	1.25984	1.29921	1.33858	1.37795	1.41732	1.45669	1.49606	1.53543
40	1.57480	1.61417	1.65354	1.69291	1.73228	1.77165	1.81102	1.85039	1.88976	1.92913
50	1.96850	2.00787	2.04724	2.08661	2.12598	2.16535	2.20472	2.24409	2.28346	2.32283
60	2.36220	2.40157	2.44094	2.48031	2.51969	2.55906	2.59843	2.63780	2.67717	2.71654
70	2.75591	2.79528	2.83465	2.87402	2.91339	2.95276	2.99213	3.03150	3.07087	3.11024
80	3.14961	3.18898	3.22835	3.26772	3.30709	3.34646	3.38583	3.42520	3.46457	3.50394
90	3.54331	3.58268	3.62205	3.66142	3.70079	3.74016	3.77953	3.81890	3.85827	3.89764
100	3.93701	3.97638	4.01575	4.05512	4.09449	4.13386	4.17323	4.21260	4.25197	4.29134
110	4.33071	4.37008	4.40945	4.44882	4.48819	4.52756	4.56693	4.60630	4.64567	4.68504
120	4.72441	4.76378	4.80315	4.84252	4.88189	4.92126	4.96063	5.00000	5.03937	5.07874
130	5.11811	5.15748	5.19685	5.23622	5.27559	5.31496	5.35433	5.39370	5.43307	5.47244
140	5.51181	5.55118	5.59055	5.62992	5.66929	5.70866	5.74803	5.78740	5.82677	5.86614
150	5.90551	5.94488	5.98425	6.02362	6.06299	6.10236	6.14173	6.18110	6.22047	6.25984
160	6.29921	6.33858	6.37795	6.41732	6.45669	6.49606	6.53543	6.57480	6.61417	6.65354
170	6.69291	6.73228	6.77165	6.81102	6.85039	6.88976	6.92913	6.96850	7.00787	7.04724
180	7.08661	7.12598	7.16535	7.20472	7.24409	7.28346	7.32283	7.36220	7.40157	7.44094
190	7.48031	7.51969	7.55906	7.59843	7.63780	7.67717	7.71654	7.75591	7.79528	7.83465
200	7.87402	7.91339	7.95276	7.99213	8.03150	8.07087	8.11024	8.14961	8.18898	8.22835
210	8.26772	8.30709	8.34646	8.38583	8.42520	8.46457	8.50394	8.54331	8.58268	8.62205
220	8.66142	8.70079	8.74016	8.77953	8.81890	8.85827	8.89764	8.93701	8.97638	9.01575
230	9.05512	9.09449	9.13386	9.17323	9.21260	9.25197	9.29134	9.33071	9.37008	9.40945
240	9.44882	9.48819	9.52756	9.56693	9.60630	9.64567	9.68504	9.72441	9.76378	9.80315
250	9.84252	9.88189	9.92126	9.96063	10.0000	10.0394	10.0787	10.1181	10.1575	10.1969
260	10.2362	10.2756	10.3150	10.3543	10.3937	10.4331	10.4724	10.5118	10.5512	10.5906
270	10.6299	10.6693	10.7087	10.7480	10.7874	10.8268	10.8661	10.9055	10.9449	10.9843
280	11.0236	11.0630	11.1024	11.1417	11.1811	11.2205	11.2598	11.2992	11.3386	11.3780
290	11.4173	11.4567	11.4961	11.5354	11.5748	11.6142	11.6535	11.6929	11.7323	11.7717
300	11.8110	11.8504	11.8898	11.9291	11.9685	12.0079	12.0472	12.0866	12.1260	12.1654
310	12.2047	12.2441	12.2835	12.3228	12.3622	12.4016	12.4409	12.4803	12.5197	12.5591
320	12.5984	12.6378	12.6772	12.7165	12.7559	12.7953	12.8346	12.8740	12.9134	12.9528
330	12.9921	13.0315	13.0709	13.1102	13.1496	13.1890	13.2283	13.2677	13.3071	13.3465
340	13.3858	13.4252	13.4646	13.5039	13.5433	13.5827	13.6220	13.6614	13.7008	13.7402
350	13.7795	13.8189	13.8583	13.8976	13.9370	13.9764	14.0157	14.0551	14.0945	14.1339
360	14.1732	14.2126	14.2520	14.2913	14.3307	14.3701	14.4094	14.4488	14.4882	14.5276
370	14.5669	14.6063	14.6457	14.6850	14.7244	14.7638	14.8031	14.8425	14.8819	14.9213
380	14.9606	15.0000	15.0394	15.0787	15.1181	15.1575	15.1969	15.2362	15.2756	15.3150
390	15.3543	15.3937	15.4331	15.4724	15.5118	15.5512	15.5906	15.6299	15.6693	15.7087
400	15.7480	15.7874	15.8268	15.8661	15.9055	15.9449	15.9843	16.0236	16.0630	16.1024
410	16.1417	16.1811	16.2205	16.2598	16.2992	16.3386	16.3780	16.4173	16.4567	16.4961
420	16.5354	16.5748	16.6142	16.6535	16.6929	16.7323	16.7717	16.8110	16.8504	16.8898
430	16.9291	16.9685	17.0079	17.0472	17.0866	17.1260	17.1654	17.2047	17.2441	17.2835
440	17.3228	17.3622	17.4016	17.4409	17.4803	17.5197	17.5591	17.5984	17.6378	17.6772
450	17.7165	17.7559	17.7953	17.8346	17.8740	17.9134	17.9528	17.9921	18.0315	18.0709
460	18.1102	18.1496	18.1890	18.2283	18.2677	18.3071	18.3465	18.3858	18.4252	18.4646
470	18.5039	18.5433	18.5827	18.6220	18.6614	18.7008	18.7402	18.7795	18.8189	18.8583
480	18.8976	18.9370	18.9764	19.0157	19.0551	19.0945	19.1339	19.1732	19.2126	19.2520
490	19.2913	19.3307	19.3701	19.4094	19.4488	19.4882	19.5276	19.5669	19.6063	19.6457
500	19.6850	19.7244	19.7638	19.8031	19.8425	19.8819	19.9213	19.9606	20.0000	20.0394
510	20.0787	20.1181	20.1575	20.1969	20.2362	20.2756	20.3150	20.3543	20.3937	20.4331
520	20.4724	20.5118	20.5512	20.5906	20.6299	20.6693	20.7087	20.7480	20.7874	20.8268
530	20.8661	20.9055	20.9449	20.9843	21.0236	21.0630	21.1024	21.1417	21.1811	21.2205
540	21.2598	21.2992	21.3386	21.3780	21.4173	21.4567	21.4961	21.5354	21.5748	21.6142
550	21.6535	21.6929	21.7323	21.7717	21.8110	21.8504	21.8898	21.9291	21.9685	22.0079
560	22.0472	22.0866	22.1260	22.1654	22.2047	22.2441	22.2835	22.3228	22.3622	22.4016
570	22.4409	22.4803	22.5197	22.5591	22.5984	22.6378	22.6772	22.7165	22.7559	22.7953
580	22.8346	22.8740	22.9134	22.9528	22.9921	23.0315	23.0709	23.1102	23.1496	23.1890
590	23.2283	23.2677	23.3071	23.3465	23.3858	23.4252	23.4646	23.5039	23.5433	23.5827
600	23.6220	23.6614	23.7008	23.7402	23.7795	23.8189	23.8583	23.8976	23.9370	23.9764
610	24.0157	24.0551	24.0945	24.1339	24.1732	24.2126	24.2520	24.2913	24.3307	24.3701
620	24.4094	24.4488	24.4882	24.5276	24.5669	24.6063	24.6457	24.6850	24.7244	24.7638
630	24.8031	24.8425	24.8819	24.9213	24.9606	25.0000	25.0394	25.0787	25.1181	25.1575
640	25.1969	25.2362	25.2756	25.3150	25.3543	25.3937	25.4331	25.4724	25.5118	25.5512
650	25.5906	25.6299	25.6693	25.7087	25.7480	25.7874	25.8268	25.8661	25.9055	25.9449
660	25.9843	26.0236	26.0630	26.1024	26.1417	26.1811	26.2205	26.2598	26.6992	26.3386
670	26.3780	26.4137	26.4567	26.4961	26.5354	26.5748	26.6142	26.6535	26.6929	26.7323
680	26.7717	26.8110	26.8504	26.8898	26.9291	26.9685	27.0079	27.0472	27.0866	27.1260
690	27.1654	27.2047	27.2441	27.2835	27.3228	27.3622	27.4016	27.4409	27.4803	27.5197
700	27.5591	27.5984	27.6378	27.6772	27.7165	27.7559	27.7953	27.8346	27.8740	27.9134
710	27.9528	27.9921	28.0315	28.0709	28.1102	28.1496	28.1890	28.2283	28.2677	28.3071
720	28.3465	28.3858	28.4252	28.4646	28.5039	28.5433	28.5827	28.6220	28.6614	28.7008
730	28.7402	28.7795	28.8189	28.8583	28.8976	28.9370	28.9764	29.0157	29.0551	29.0945
740	29.1339	29.1732	29.2126	29.2520	29.2913	29.3307	29.3701	29.4094	29.4488	29.4882

Temperature conversion chart

Celsius to Fahrenheit

Note: The numbers in boldface refer to temperature in degrees, either Celsius or Fahrenheit, which it is desired to convert to the other scale. If converting from Fahrenheit to Celsius degrees, the equivalent temperature will be found in the left column; while if converting from Celsius to Fahrenheit, the answer will be found in the column on the right.

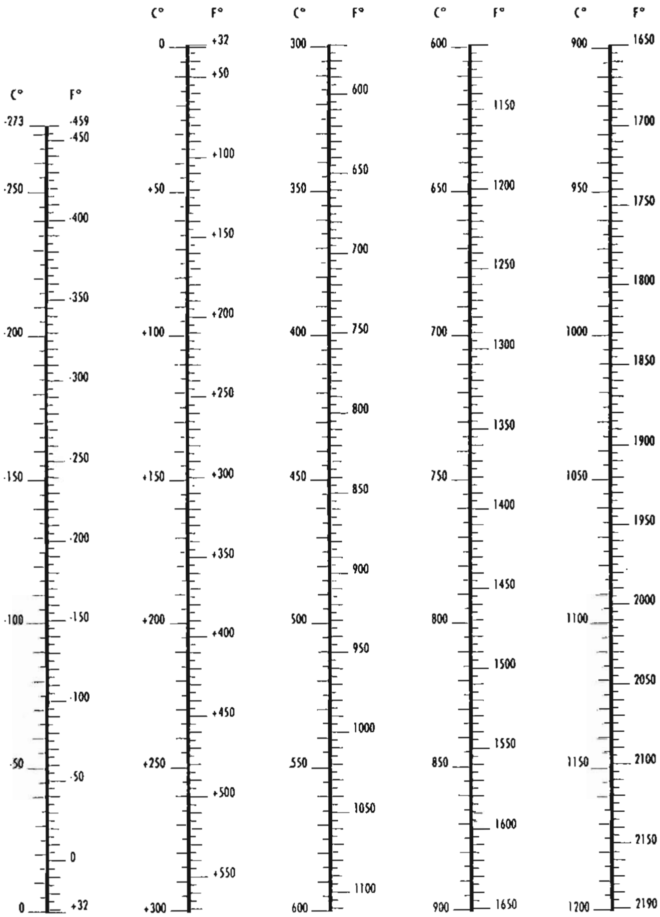
Celsius		Fahrenheit	Celsius		Fahrenheit	Celsius		Fahrenheit	Celsius		Fahrenheit
-73.3	**-100**	-148.0	2.8	**37**	98.6	33.3	**92**	197.6	293	**560**	1040.0
-67.8	**-90**	-130.0	3.3	**38**	100.4	33.9	**93**	199.4	299	**570**	1058.0
-62.2	**-80**	-112.0	3.9	**39**	102.2	34.4	**94**	201.2	304	**580**	1076.0
-59.4	**-75**	-103.0	4.4	**40**	104.0	35.0	**95**	203.0	310	**590**	1094.0
-56.7	**-70**	-94.0	5.0	**41**	105.8	35.6	**96**	204.8	316	**600**	1112.0
-53.9	**-65**	-85.0	5.6	**42**	107.6	36.1	**97**	206.6	321	**610**	1130.0
-51.1	**-60**	-76.0	6.1	**43**	109.4	36.7	**98**	208.4	327	**620**	1148.0
-48.3	**-55**	-67.0	6.7	**44**	111.2	37.2	**99**	210.2	332	**630**	1166.0
-45.6	**-50**	-58.0	7.2	**45**	113.0	37.8	**100**	212.0	338	**640**	1184.0
-42.8	**-45**	-49.0	7.8	**46**	114.8	43.0	**110**	230.0	343	**650**	1202.0
-40.0	**-40**	-40.0	8.3	**47**	116.6	49.0	**120**	248.0	349	**660**	1220.0
-37.2	**-35**	-31.0	8.9	**48**	118.4	54.0	**130**	266.0	354	**670**	1238.0
-34.4	**-30**	-22.0	9.4	**49**	120.2	60.0	**140**	284.0	360	**680**	1256.0
-31.7	**-25**	-13.0	10.0	**50**	122.0	66.0	**150**	302.0	366	**690**	1274.0
-28.9	**-20**	-4.0	10.6	**51**	123.8	71.0	**160**	320.0	371	**700**	1292.0
-26.1	**-15**	5.0	11.1	**52**	125.6	77.0	**170**	338.0	377	**710**	1310.0
-23.3	**-10**	14.0	11.7	**53**	127.4	82.0	**180**	356.0	382	**720**	1328.0
-20.6	**-5**	23.0	12.2	**54**	129.2	88.0	**190**	374.0	388	**730**	1346.0
-17.8	**0**	32.0	12.8	**55**	131.0	93.0	**200**	392.0	393	**740**	1364.0
-17.2	**1**	33.8	13.3	**56**	132.8	99.0	**210**	410.0	399	**750**	1382.0
-16.7	**2**	35.6	13.9	**57**	134.6	100.0	**212**	414.0	404	**760**	1400.0
-16.1	**3**	37.4	14.4	**58**	136.4	104.0	**220**	428.0	410	**770**	1418.0
-15.6	**4**	39.2	15.0	**59**	138.2	110.0	**230**	446.0	416	**780**	1436.0
-15.0	**5**	41.0	15.6	**60**	140.0	116.0	**240**	464.0	421	**790**	1454.0
-14.4	**6**	42.8	16.1	**61**	141.8	121.0	**250**	482.0	427	**800**	1472.0
-13.9	**7**	44.6	16.7	**62**	143.6	127.0	**260**	500.0	432	**810**	1490.0
-13.3	**8**	46.4	17.2	**63**	145.4	132.0	**270**	518.0	438	**820**	1508.0
-12.8	**9**	48.2	17.8	**64**	147.2	138.0	**280**	536.0	443	**830**	1526.0
-12.2	**10**	50.0	18.3	**65**	149.0	143.0	**290**	554.0	449	**840**	1544.0
-11.7	**11**	51.8	18.9	**66**	150.8	149.0	**300**	572.0	454	**850**	1562.0
-11.1	**12**	53.6	19.4	**67**	152.6	154.0	**310**	590.0	460	**860**	1580.0
-10.6	**13**	55.4	20.0	**68**	154.4	160.0	**320**	608.0	466	**870**	1598.0
-10.0	**14**	57.2	20.6	**69**	156.2	166.0	**330**	626.0	471	**880**	1616.0
-9.4	**15**	59.0	21.1	**70**	158.0	171.0	**340**	644.0	477	**890**	1634.0
-8.9	**16**	60.8	21.7	**71**	159.8	177.0	**350**	662.0	482	**900**	1652.0
-8.3	**17**	62.6	22.2	**72**	161.6	182.0	**360**	680.0	488	**910**	1670.0
-7.8	**18**	64.4	22.8	**73**	163.4	188.0	**370**	698.0	493	**920**	1688.0
-7.2	**19**	66.2	23.3	**74**	165.2	193.0	**380**	716.0	499	**930**	1706.0
-6.7	**20**	68.0	23.9	**75**	167.0	199.0	**390**	734.0	504	**940**	1724.0
-6.1	**21**	69.8	24.4	**76**	168.8	204.0	**400**	752.0	510	**950**	1742.0
-5.6	**22**	71.6	25.0	**77**	170.6	210.0	**410**	770.0	516	**960**	1760.0
-5.0	**23**	73.4	25.6	**78**	172.4	216.0	**420**	788.0	521	**970**	1778.0
-4.4	**24**	75.2	26.1	**79**	174.2	221.0	**430**	806.0	527	**980**	1796.0
-3.9	**25**	77.0	26.7	**80**	176.0	227.0	**440**	824.0	532	**990**	1814.0
-3.3	**26**	78.8	27.2	**81**	177.8	232.0	**450**	842.0	538	**1000**	1832.0
-2.8	**27**	80.6	27.8	**82**	179.6	238.0	**460**	860.0	566	**1050**	1922.0
-2.2	**28**	82.4	28.3	**83**	181.4	243.0	**470**	878.0	593	**1100**	2012.0
-1.7	**29**	84.2	28.9	**84**	183.2	249.0	**480**	896.0	621	**1150**	2102.0
-1.1	**30**	86.0	29.4	**85**	185.0	254.0	**490**	914.0	649	**1200**	2192.0
-0.6	**31**	87.8	30.0	**86**	186.8	260.0	**500**	932.0	677	**1250**	2282.0
0.0	**32**	89.6	30.6	**87**	188.6	266.0	**510**	950.0	704	**1300**	2372.0
0.6	**33**	91.4	31.1	**88**	190.4	271.0	**520**	968.0	732	**1350**	2462.0
1.1	**34**	93.2	31.7	**89**	192.2	277.0	**530**	986.0	760	**1400**	2552.0
1.7	**35**	95.0	32.2	**90**	194.0	282.0	**540**	1004.0	788	**1450**	2642.0
2.2	**36**	96.8	32.8	**91**	195.8	288.0	**550**	1022.0	816	**1500**	2732.0

The formulae on the right may also be used for converting Celsius or Fahrenheit degrees into the other scale.

Degrees Cels. $°C = \frac{5}{9}(°F + 40) - 40$ Degrees Fahr. $°F = \frac{9}{5}(°C + 40) - 40$

Temperature

$$°F = 9/5\ °C + 32$$
$$°C = 5/9\ (°F - 32)$$



Pressure conversions

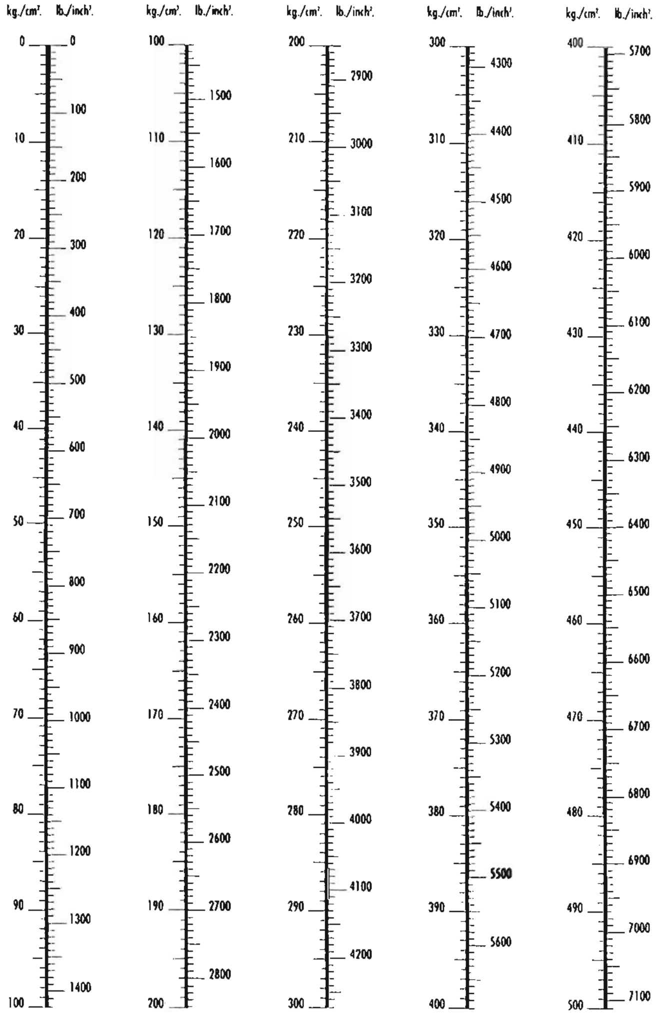
Pounds per square inch (lbf/in^2) to bar

1–40		41–80		81–200		205–500		510–900		910–1500	
lbf/in^2	bar	lbf/in^2	bar	lbf/in^2	bar	lbf/in^2	bar	lbf/in^2	bar	lbf/in^2	bar
1	0.07	41	2.83	81	5.58	205	14.13	510	35.16	910	62.74
2	0.14	42	2.90	82	5.65	210	14.48	520	35.85	920	63.43
3	0.21	43	2.96	83	5.72	215	14.82	530	36.54	930	64.12
4	0.28	44	3.03	84	5.79	220	15.17	540	37.23	940	64.81
5	0.34	45	3.10	85	5.86	225	15.51	550	37.92	950	65.50
6	0.41	46	3.17	86	5.93	230	15.86	560	38.61	960	66.19
7	0.48	47	3.24	87	6.00	235	16.20	570	39.30	970	66.88
8	0.55	48	3.31	88	6.07	240	16.55	580	39.99	980	67.57
9	0.62	49	3.38	89	6.14	245	16.89	590	40.68	990	68.26
10	0.69	50	3.45	90	6.21	250	17.24	600	41.37	1000	68.95
11	0.76	51	3.52	91	6.27	255	17.58	610	42.06	1010	69.64
12	0.83	52	3.59	92	6.34	260	17.93	620	42.75	1020	70.33
13	0.90	53	3.65	93	6.41	265	18.27	630	43.44	1030	71.02
14	0.97	54	3.72	94	6.48	270	18.62	640	44.13	1040	71.71
15	1.03	55	3.79	95	6.55	275	18.96	650	44.82	1050	72.39
16	1.10	56	3.86	96	6.62	280	19.31	660	45.51	1060	73.08
17	1.17	57	3.93	97	6.69	285	19.65	670	46.19	1070	73.77
18	1.24	58	4.00	98	6.76	290	19.99	680	46.88	1080	74.46
19	1.31	59	4.07	99	6.83	295	20.34	690	47.57	1090	75.15
20	1.38	60	4.14	100	6.89	300	20.68	700	48.26	1100	75.84
21	1.45	61	4.21	105	7.24	310	21.37	710	48.95	1120	77.22
22	1.52	62	4.27	110	7.58	320	22.06	720	49.64	1140	78.60
23	1.59	63	4.34	115	7.93	330	22.75	730	50.33	1160	79.98
24	1.65	64	4.41	120	8.27	340	23.44	740	51.02	1180	81.36
25	1.72	65	4.48	125	8.62	350	24.13	750	51.71	1200	82.74
26	1.79	66	4.55	130	8.96	360	24.82	760	52.40	1220	84.12
27	1.86	67	4.62	135	9.31	370	25.51	770	53.09	1240	85.49
28	1.93	68	4.69	140	9.65	380	26.20	780	53.78	1260	86.87
29	2.00	69	4.76	145	10.00	390	26.89	790	54.47	1280	88.25
30	2.07	70	4.83	150	10.34	400	27.58	800	55.16	1300	89.63
31	2.14	71	4.90	155	10.69	410	28.27	810	55.85	1320	91.01
32	2.21	72	4.96	160	11.03	420	28.96	820	56.54	1340	92.39
33	2.28	73	5.03	165	11.38	430	29.65	830	57.23	1360	93.77
34	2.34	74	5.10	170	11.72	440	30.34	840	57.92	1380	95.15
35	2.41	75	5.17	175	12.07	450	31.03	850	58.61	1400	96.53
36	2.48	76	5.24	180	12.41	460	31.72	860	59.29	1420	97.91
37	2.55	77	5.31	185	12.76	470	32.41	870	59.98	1440	99.28
38	2.62	78	5.38	190	13.10	480	33.09	880	60.67	1460	100.66
39	2.69	79	5.45	195	13.44	490	33.78	890	61.36	1480	102.04
40	2.76	80	5.52	200	13.79	500	34.47	900	62.05	1500	103.42

Pressure

1 Kg/cm² = 142233 lb/inch²

1 lb/inch² = 0,07037 Kg/cm²



Steam tables

Metric SI units

Pressure			Temperature	Specific enthalpy			Specific volume
bar		kPa	°C	Water (h_f) kJ/kg	Evaporation (h_{fg}) kJ/kg	Steam (h_g) kJ/kg	steam (v_g) m^3/kg
0.30		30.0	69.10	289.23	2336.1	2625.3	5.229
0.50	absolute	50.0	81.33	340.49	2305.4	2645.9	3.240
0.75		75.0	91.78	384.39	2278.6	2663.0	2.217
0.95		95.0	98.20	411.43	2261.8	2673.2	1.777
0	gauge	0	100.00	419.04	2257.0	2676.0	1.673
0.10		10.0	102.66	430.2	2250.2	2680.4	1.533
0.20		20.0	105.10	440.8	2243.4	2684.2	1.414
0.30		30.0	107.39	450.4	2237.2	2687.6	1.312
0.40		40.0	109.55	459.7	2231.3	2691.0	1.225
0.50		50.0	111.61	468.3	2225.6	2693.9	1.149
0.60		60.0	113.56	476.4	2220.4	2696.8	1.083
0.70		70.0	115.40	484.1	2215.4	2699.5	1.024
0.80		80.0	117.14	491.6	2210.5	2702.1	0.971
0.90		90.0	118.80	498.9	2205.6	2704.5	0.923
1.00		100.0	120.42	505.6	2201.1	2706.7	0.881
1.10		110.0	121.96	512.2	2197.0	2709.2	0.841
1.20		120.0	123.46	518.7	2192.8	2711.5	0.806
1.30		130.0	124.90	524.6	2188.7	2713.3	0.773
1.40		140.0	126.28	530.5	2184.8	2715.3	0.743
1.50		150.0	127.62	536.1	2181.0	2717.1	0.714
1.60		160.0	128.89	541.6	2177.3	2718.9	0.689
1.70		170.0	130.13	547.1	2173.7	2720.8	0.665
1.80		180.0	131.37	552.3	2170.1	2722.4	0.643
1.90		190.0	132.54	557.3	2166.7	2724.0	0.622
2.00		200.0	133.69	562.2	2163.3	2725.5	0.603
2.20		220.0	135.88	571.7	2156.9	2728.6	0.568
2.40		240.0	138.01	580.7	2150.7	2731.4	0.536
2.60		260.0	140.00	589.2	2144.7	2733.9	0.509
2.80		280.0	141.92	597.4	2139.0	2736.4	0.483
3.00		300.0	143.75	605.3	2133.4	2738.7	0.461
3.20		320.0	145.46	612.9	2128.1	2741.0	0.440
3.40		340.0	147.20	620.0	2122.9	2742.9	0.422
3.60		360.0	148.84	627.1	2117.8	2744.9	0.405
3.80		380.0	150.44	634.0	2112.9	2746.9	0.389
4.00		400.0	151.96	640.7	2108.1	2748.8	0.374
4.50		450.0	155.55	655.3	2096.7	2753.0	0.342
5.00		500.0	158.92	670.9	2086.0	2756.9	0.315
5.50		550.0	162.08	684.6	2075.7	2760.3	0.292
6.00		600.0	165.04	697.5	2066.0	2763.5	0.272
6.50		650.0	167.83	709.7	2056.8	2766.5	0.255
7.00		700.0	170.50	721.4	2047.7	2769.1	0.240
7.50		750.0	173.02	732.5	2039.2	2771.7	0.227
8.00		800.0	175.43	743.1	2030.9	2774.0	0.215
8.50		850.0	177.75	753.3	2022.9	2776.2	0.204

Steam tables

Metric SI units

Pressure		Temperature	Specific enthalpy			Specific volume
			Water (h_f)	Evaporation (h_{fg})	Steam (h_g)	steam (v_g)
bar	kPa	°C	kJ/kg	kJ/kg	kJ/kg	m^3/kg
9.00	900.0	179.97	763.0	2015.1	2778.1	0.194
9.50	950.0	182.10	772.5	2007.5	2780.0	0.185
10.00	1000.0	184.13	781.6	2000.1	2781.7	0.177
10.50	1050.0	186.05	790.1	1993.0	2783.3	0.171
11.00	1100.0	188.02	798.8	1986.0	2784.8	0.163
11.50	1150.0	189.82	807.1	1979.1	2786.3	0.157
12.00	1200.0	191.68	815.1	1972.5	2787.6	0.151
12.50	1250.0	193.43	822.9	1965.4	2788.8	0.148
13.00	1300.0	195.10	830.4	1959.6	2790.0	0.141
13.50	1350.0	196.62	837.9	1953.2	2791.1	0.136
14.00	1400.0	198.35	845.1	1947.1	2792.2	0.132
14.50	1450.0	199.92	852.1	1941.0	2793.1	0.128
15.00	1500.0	201.45	859.0	1935.0	2794.0	0.124
16.00	1600.0	204.38	872.3	1923.4	2795.7	0.117
17.00	1700.0	207.17	885.0	1912.1	2797.1	0.110
18.00	1800.0	209.90	897.2	1901.3	2798.5	0.105
19.00	1900.0	212.47	909.0	1890.5	2799.5	0.100
20.00	2000.0	214.96	920.3	1880.2	2800.5	0.0949
21.00	2100.0	217.35	931.3	1870.1	2801.4	0.0906
22.00	2200.0	219.65	941.9	1860.1	2802.0	0.0868
23.00	2300.0	221.85	952.2	1850.4	2802.6	0.0832
24.00	2400.0	224.02	962.2	1804.9	2803.1	0.0797
25.00	2500.0	226.12	972.1	1831.4	2803.5	0.0768
26.00	2600.0	228.15	981.6	1822.2	2803.8	0.0740
27.00	2700.0	230.14	990.7	1813.3	2804.0	0.0714
28.00	2800.0	232.05	999.7	1804.4	2804.1	0.0689
29.00	2900.0	233.93	1008.6	1795.6	2804.2	0.0666
30.00	3000.0	235.78	1017.0	1787.0	2804.1	0.0645
31.00	3100.0	237.55	1025.6	1778.5	2804.1	0.0625
32.00	3200.0	239.28	1033.9	1770.0	2803.9	0.0605
33.00	3300.0	240.97	1041.9	1761.8	2803.7	0.0587
34.00	3400.0	242.63	1049.7	1753.8	2803.5	0.0571
35.00	3500.0	244.26	1057.7	1745.5	2803.2	0.0554
36.00	3600.0	245.86	1065.7	1737.2	2802.9	0.0539
37.00	3700.0	247.42	1072.9	1729.5	2802.4	0.0524
38.00	3800.0	248.95	1080.3	1721.6	2801.9	0.0510
39.00	3900.0	250.42	1087.4	1714.1	2801.5	0.0498
40.00	4000.0	251.94	1094.6	1706.3	2800.9	0.0485
42.00	4200.0	254.74	1108.6	1691.2	2799.8	0.0461
44.00	4400.0	257.50	1122.1	1676.2	2798.2	0.0441
46.00	4600.0	260.13	1135.3	1661.6	2796.9	0.0421
48.00	4800.0	262.73	1148.1	1647.1	2795.2	0.0403
50.00	5000.0	265.26	1160.8	1632.8	2793.6	0.0386

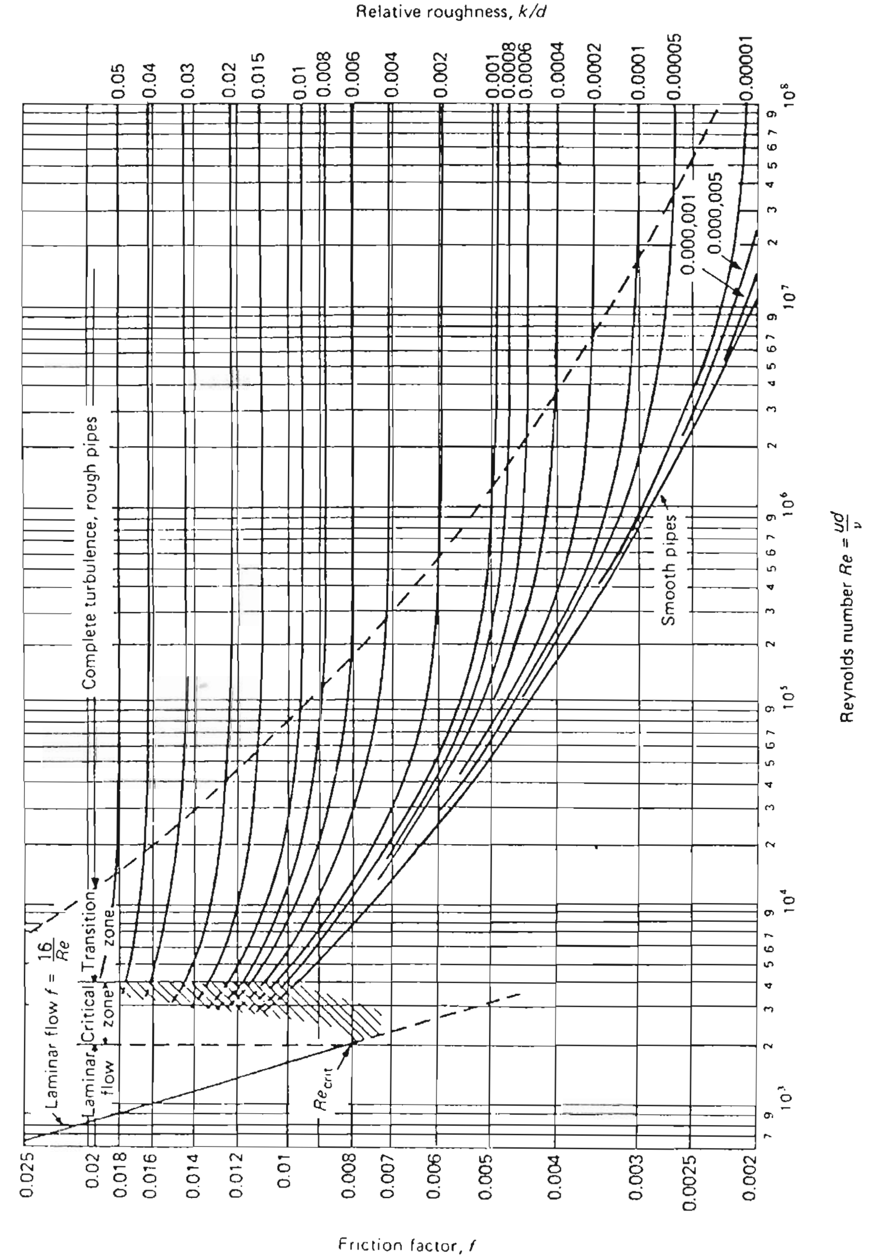
Imperial units

Pressure		Temperature °F	Sensible heat Btu/lb	Latent heat Btu/lb	Total heat Btu/lb	Volume dry saturation cu ft/lb
Inches of vacuum	15	179.0	147.0	991.0	1138.0	51.41
	10	192.0	160.0	983.0	1143.0	39.40
	5	203.0	171.0	976.0	1147.0	31.80
psig	0	212.0	180.2	970.6	1150.8	26.80
	2	218.5	186.8	966.4	1153.2	23.80
	4	224.5	192.7	962.6	1155.3	21.40
	6	230.0	198.1	959.2	1157.3	19.40
	8	234.8	203.1	956.0	1159.1	17.90
	10	239.4	207.9	952.9	1160.8	16.50
	12	243.7	212.3	950.1	1162.3	15.30
	14	247.9	216.4	947.3	1163.7	14.30
	16	251.7	220.3	944.8	1165.1	13.40
	18	255.4	224.0	942.4	1166.4	12.70
	20	258.8	227.5	940.1	1167.6	12.00
	22	262.3	230.9	937.8	1168.7	11.40
	24	265.3	234.2	935.8	1170.0	10.80
	26	268.3	273.3	933.5	1170.8	10.30
	28	271.4	240.2	931.6	1171.8	9.87
	30	274.0	243.0	929.7	1172.7	9.46
	32	276.7	245.9	927.6	1173.5	9.08
	34	279.4	248.5	925.8	1174.3	8.73
	36	281.9	251.1	924.0	1175.1	8.40
	38	284.4	253.7	922.1	1175.8	8.11
	40	286.7	256.1	920.4	1176.5	7.83
	42	289.0	258.5	918.6	1177.1	7.57
	44	291.3	260.8	917.0	1177.8	7.33
	46	293.5	263.0	915.4	1178.4	7.10
	48	295.6	265.2	913.8	1179.0	6.89
	50	297.7	267.4	912.2	1179.6	6.68
	52	299.7	269.4	910.7	1180.1	6.50
	54	301.7	271.5	909.2	1180.7	6.32
	56	303.6	273.5	907.8	1181.3	6.16
	58	305.5	275.3	906.5	1181.8	6.00
	60	307.4	277.1	905.3	1182.4	5.84
	62	309.2	279.0	904.0	1183.0	5.70
	64	310.9	280.9	902.6	1183.5	5.56
	66	312.7	282.8	901.2	1184.0	5.43
	68	314.3	284.5	900.0	1184.5	5.31
	70	316.0	286.2	898.8	1185.0	5.19
	72	317.7	288.0	897.5	1185.5	5.08
	74	319.3	289.4	896.5	1185.9	4.97
	76	320.9	291.2	895.1	1186.3	4.87
	78	322.4	292.9	893.9	1186.8	4.77
	80	323.9	294.5	892.7	1187.2	4.67
	82	325.5	296.1	891.5	1187.6	4.58
	84	326.9	297.6	890.3	1187.9	4.49

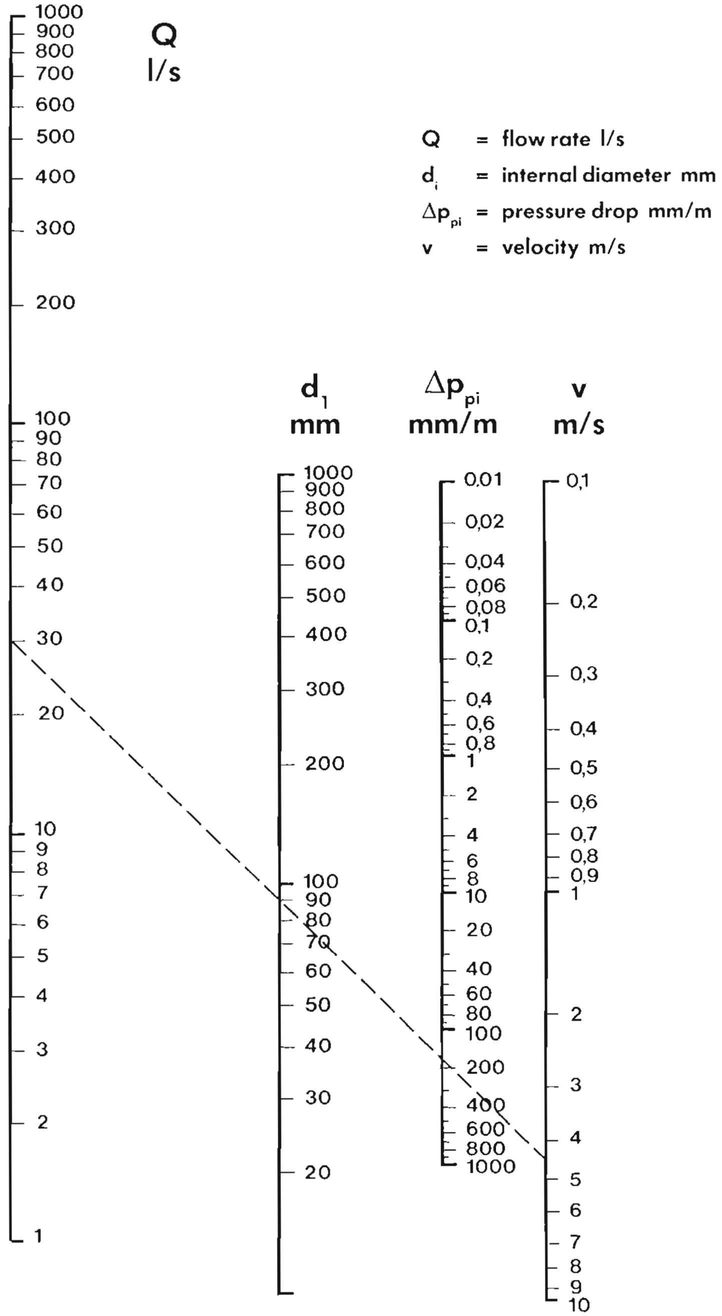
Imperial units

Pressure	Temperature °F	Sensible heat Btu/lb	Latent heat Btu/lb	Total heat Btu/lb	Volume dry saturation cu ft/lb
86	328.4	299.1	889.2	1188.3	4.41
88	329.9	300.6	888.1	1188.7	4.33
90	331.2	302.1	887.0	1189.1	4.25
92	332.6	303.5	885.8	1189.3	4.17
94	333.9	304.9	884.8	1189.7	4.10
96	335.3	306.3	883.7	1190.0	4.03
98	336.6	307.7	882.6	1190.3	3.96
100	337.9	309.0	881.6	1190.6	3.90
105	341.1	312.3	879.0	1191.4	3.74
110	344.2	315.5	876.5	1192.0	3.60
115	347.1	318.7	874.0	1192.7	3.46
120	350.1	321.8	871.5	1193.3	3.34
125	352.8	324.7	869.3	1194.0	3.23
130	355.6	327.6	866.9	1194.5	3.12
135	358.3	330.6	864.5	1195.1	3.02
140	360.9	333.2	862.5	1195.7	2.93
145	363.5	335.9	860.3	1196.2	2.84
150	365.9	338.6	858.0	1196.6	2.76
160	370.7	343.6	853.9	1197.5	2.61
170	375.2	348.5	849.8	1198.3	2.48
180	379.6	353.2	845.9	1199.1	2.35
190	383.7	357.6	842.2	1199.8	2.24
200	387.7	362.0	838.4	1200.4	2.14
210	391.7	366.2	834.8	1201.0	2.04
220	395.5	370.3	831.2	1201.5	1.96
230	399.1	374.2	827.8	1202.0	1.88
240	402.7	378.0	824.5	1202.5	1.81
250	406.1	381.7	821.2	1202.9	1.74
260	409.3	385.3	817.9	1203.2	1.68
270	412.5	388.8	814.8	1203.6	1.62
280	415.8	392.3	811.6	1203.9	1.57
290	418.8	395.7	808.5	1204.2	1.52
300	421.7	398.9	805.5	1204.4	1.47
310	424.7	402.1	802.6	1204.7	1.43
320	427.5	405.2	799.7	1204.9	1.39
330	430.3	408.3	796.7	1205.0	1.35
340	433.0	411.3	793.8	1205.1	1.31
350	435.7	414.3	791.0	1205.3	1.27
360	438.3	417.2	788.2	1205.4	1.24
370	440.8	420.0	785.4	1205.4	1.21
380	443.3	422.8	782.7	1205.5	1.18
390	445.7	425.6	779.9	1205.5	1.15
400	448.1	428.2	777.4	1205.6	1.12

Moody's diagram



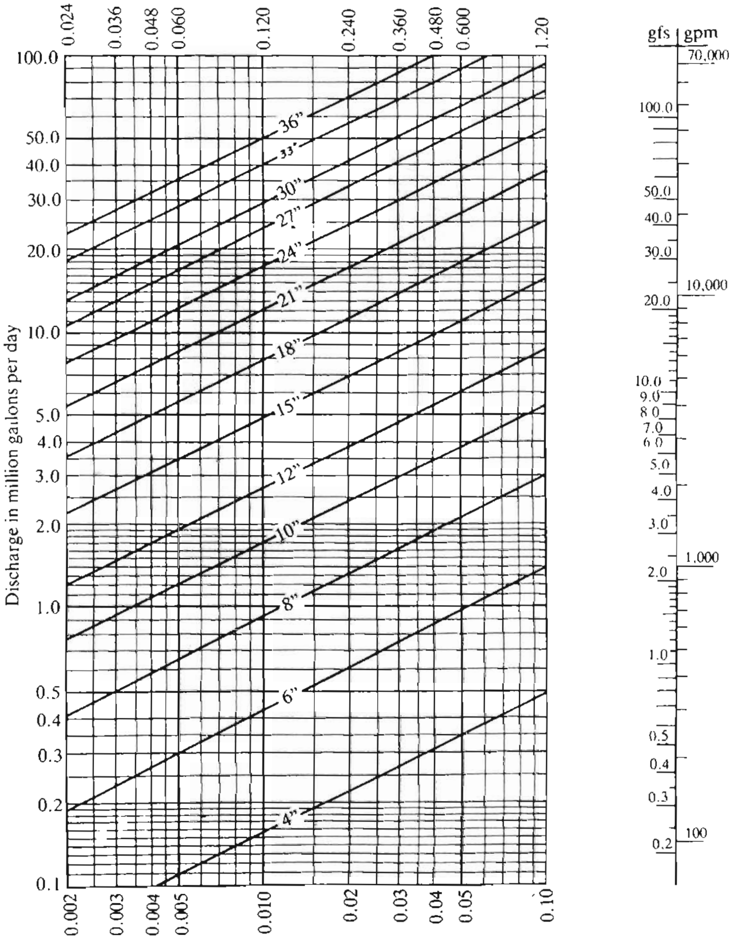
Nomogram for pipe losses



Nomograph solution of Manning's formula for discharge of circular pipes running full (n = 0.011)

$$Q = A \frac{1.486}{n} R^{2/3} S^{1/2}$$

where, Q is the discharge in millions of gallons per day
A is the area of wetted cross-section of the pipe in square feet
n is the empirically derived coefficient used to represent the interior surface characteristics of the pipe
R is the hydraulic radius of wetted cross-section of the pipe in feet
S is the slope of hydraulic gradient



Slope of hydraulic gradient
n = 0.011

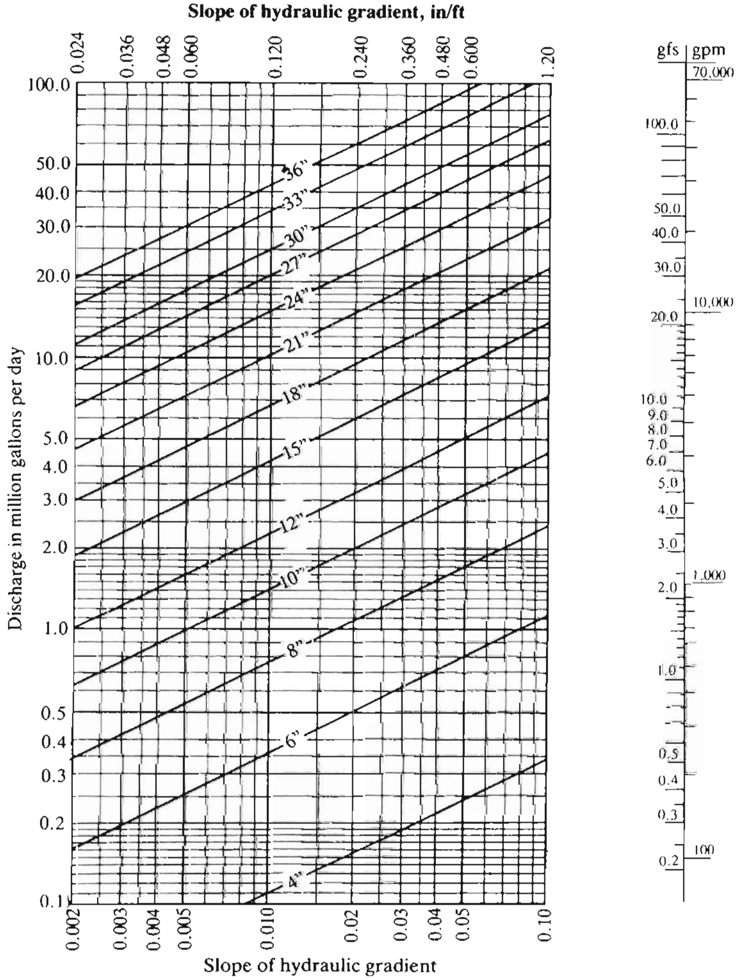
Notes:

1. Unless otherwise known, a value of 0.013 is recommended for n for pipes of all materials.
2. The velocity of flow (averaged over the wetted cross-section) should be kept between 2 ft/sec and 10 ft/sec.

Nomograph solution of Manning's formula for discharge of circular pipes running full (n = 0.013)

$$Q = A \frac{1.486}{n} R^{2/3} S^{1/2}$$

where, Q is the discharge in millions of gallons per day
A is the area of wetted cross-section of the pipe in square feet
n is the empirically derived coefficient used to represent the interior surface characteristics of the pipe
R is the hydraulic radius of wetted cross-section of the pipe in feet
S is the slope of hydraulic gradient



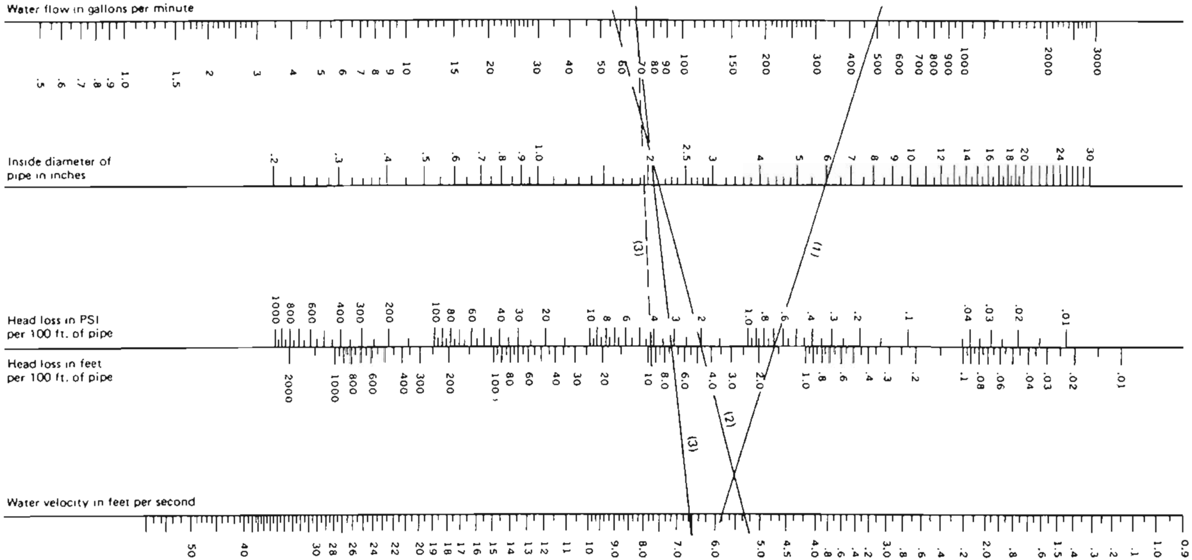
Notes:

1. Unless otherwise known, a value of 0.013 is recommended for n for pipes of all materials.
2. The velocity of flow (averaged over the wetted cross-section) should be kept between 2 ft/sec and 10 ft/sec.

Head-loss characteristics of water flow through rigid plastic pipe

This nomograph provides approximate values for a wide range of plastic pipe sizes. More precise values should be calculated from the Williams and Hazen formula. Experimental test value of C (a constant for inside pipe roughness) ranges from 155 to 165 for various types of plastic pipe. Use of a value of 150 will ensure conservative friction-loss values.

Values for head loss on PVC and CPVC fittings and valves are not available at the present time. Since directional changes and restrictions contribute the most head loss, use of head-loss data for comparable metal valves and fittings will provide conservative values.



THE VALUES OF THIS GRAPH ARE BASED ON THE WILLIAMS & HAZEN FORMULA

$$f = .2083 \left(\frac{100}{C}\right)^{1.852} \times \frac{q^{1.852}}{d^{4.8655}}$$

WHERE: f = Friction head in feet of water per 100 feet.
d = Inside diameter of pipe in inches.
q = Flowing gallons per minute

The nomograph is used by lining up values on the scale by means of a ruler or straight edge. Two independent variables must be set to obtain the other values. For example, line (1) indicates that 500 gallons per minute may be obtained with a 6-in inside diameter pipe at a head loss of about 0.65 pounds per square inch at a velocity of 6.0 feet per second. Line (2) indicates that a pipe with a 2.1-in inside diameter will give a flow of about 60 gallons per minute at a loss in head of 2 pounds per square inch per 100 feet of pipe. Line (3) and dotted line (3) show that in going from a pipe 2.1-in inside diameter to one of 2-in inside diameter the head loss goes from 3 to 4 pounds per square inch in obtaining a flow of 70 gallons per minute. Our raw material supplier does not recommend velocities in excess of 5.0 feet per second.

Nomograph courtesy of Plastics Pipe Institute, a division of The Society of the Plastics Industry.

Weight (mass) density and specific volume of gases

The mass density (ρ) of a gas or vapour in lb/ft^3 can be calculated from the following equations:

$$\rho = \frac{144\bar{P}}{RT}$$

where $\bar{P}$ = absolute pressure in lb/in^2
(= gauge pressure + 14.7)
R = individual gas constant
T = absolute temperature in °Rankine

$$\rho = \frac{M\bar{P}}{10.72T}$$

where M = molecular weight

$$\rho = \frac{270\bar{P} \times Sg}{T}$$

where Sg is the pecific gravity of the gas or vapour.

Table 1 gives the weight (mass) density for air at various temperatures and *gauge* pressures. It can also be used directly for steam. For other gases, multiply the corresponding mass density figure for air by the specific gravity of the gas. For example, for the mass density of butane at similar temperatures and pressures, factor table values by 2.067.

Table 1. Weight (mass) density of air lb/ft^3 for gauge pressures in lb/in^2

Air temperature °F	lb/in^2 0	5	10	20	30	40	50	60	70	80	90	100	110	120	130
30	0.0811	0.1087	0.1363	0.1915	0.247	0.302	0.357	0.412	0.467	0.522	0.578	0.633	0.688	0.743	0.798
40	0.0795	0.1065	0.1335	0.1876	0.242	0.295	0.350	0.404	0.458	0.512	0.566	0.620	0.674	0.728	0.782
50	0.0782	0.1048	0.1314	0.1846	0.238	0.291	0.344	0.397	0.451	0.504	0.557	0.610	0.663	0.717	0.770
60	0.0764	0.1024	0.1284	0.1804	0.232	0.284	0.336	0.388	0.440	0.492	0.544	0.596	0.648	0.700	0.752
70	0.0750	0.1005	0.1260	0.1770	0.228	0.279	0.330	0.381	0.432	0.483	0.534	0.585	0.636	0.687	0.738
80	0.0736	0.0986	0.1236	0.1737	0.224	0.274	0.324	0.374	0.424	0.474	0.524	0.574	0.624	0.674	0.724
90	0.0722	0.0968	0.1214	0.1705	0.220	0.269	0.318	0.367	0.416	0.465	0.515	0.564	0.613	0.662	0.711
100	0.0709	0.0951	0.1192	0.1675	0.216	0.264	0.312	0.361	0.409	0.457	0.505	0.554	0.602	0.650	0.698
110	0.0697	0.0934	0.1171	0.1645	0.212	0.259	0.307	0.354	0.402	0.449	0.497	0.544	0.591	0.639	0.686
120	0.0685	0.0918	0.1151	0.1617	0.208	0.255	0.302	0.348	0.395	0.441	0.488	0.535	0.581	0.628	0.674
130	0.0673	0.0902	0.1131	0.1590	0.205	0.251	0.296	0.342	0.388	0.434	0.480	0.525	0.571	0.617	0.663
140	0.0662	0.0887	0.1113	0.1563	0.201	0.246	0.291	0.337	0.382	0.427	0.472	0.517	0.562	0.607	0.652
150	0.0651	0.0873	0.1094	0.1537	0.1981	0.242	0.287	0.331	0.375	0.420	0.464	0.508	0.553	0.597	0.641
175	0.0626	0.0834	0.1051	0.1477	0.1903	0.233	0.275	0.318	0.361	0.403	0.446	0.488	0.531	0.573	0.616
200	0.0602	0.0807	0.1011	0.1421	0.1831	0.224	0.265	0.306	0.347	0.388	0.429	0.470	0.511	0.552	0.593
225	0.0580	0.0777	0.0974	0.1369	0.1764	0.216	0.255	0.295	0.334	0.374	0.413	0.453	0.492	0.531	0.571
250	0.0559	0.0750	0.0940	0.1321	0.1702	0.208	0.246	0.284	0.322	0.361	0.399	0.437	0.475	0.513	0.551
275	0.0540	0.0724	0.0908	0.1276	0.1644	0.201	0.238	0.275	0.311	0.348	0.385	0.422	0.459	0.495	0.532
300	0.0523	0.0700	0.0878	0.1234	0.1590	0.1945	0.230	0.266	0.301	0.337	0.372	0.408	0.443	0.479	0.515
350	0.0490	0.0657	0.0824	0.1158	0.1491	0.1825	0.216	0.249	0.283	0.316	0.349	0.383	0.416	0.449	0.483
400	0.0462	0.0619	0.0776	0.1090	0.1405	0.1719	0.203	0.235	0.266	0.298	0.329	0.360	0.392	0.423	0.455
450	0.0436	0.0585	0.0733	0.1030	0.1327	0.1624	0.1921	0.222	0.252	0.281	0.311	0.341	0.370	0.400	0.430
500	0.0414	0.0555	0.0695	0.0977	0.1258	0.1540	0.1821	0.210	0.238	0.267	0.295	0.323	0.351	0.379	0.407
550	0.0393	0.0527	0.0661	0.0928	0.1196	0.1464	0.1731	0.1999	0.227	0.253	0.280	0.307	0.334	0.360	0.387
600	0.0375	0.0502	0.0630	0.0885	0.1140	0.1395	0.1649	0.1904	0.216	0.241	0.267	0.292	0.318	0.343	0.369

continued

Table 1. —*continued*

Air temperature	lb/in^2													
°F	140	150	175	200	225	250	300	400	500	600	700	800	900	1000
30	0.853	0.909	1.047	1.185	1.323	1.460	1.736	2.29	2.84	3.39	3.94	4.49	5.05	5.60
40	0.836	0.890	1.026	1.161	1.296	1.431	1.702	2.24	2.78	3.32	3.86	4.40	4.95	5.49
50	0.823	0.876	1.009	1.142	1.275	1.408	1.674	2.21	2.74	3.27	3.80	4.33	4.87	5.40
60	0.804	0.856	0.986	1.116	1.246	1.376	1.636	2.16	2.68	3.20	3.72	4.24	4.76	5.28
70	0.789	0.840	0.968	1.095	1.223	1.350	1.605	2.12	2.63	3.14	3.65	4.16	4.67	5.18
80	0.774	0.824	0.950	1.075	1.200	1.325	1.575	2.08	2.58	3.08	3.58	4.08	4.58	5.08
90	0.760	0.809	0.932	1.055	1.178	1.301	1.547	2.04	2.53	3.02	3.51	4.00	4.50	4.99
100	0.747	0.795	0.916	1.036	1.157	1.278	1.519	2.00	2.48	2.97	3.45	3.93	4.42	4.90
110	0.734	0.781	0.900	1.018	1.137	1.255	1.492	1.967	2.44	2.92	3.39	3.86	4.34	4.81
120	0.721	0.768	0.884	1.001	1.117	1.234	1.467	1.933	2.40	2.86	3.33	3.80	4.26	4.73
130	0.709	0.755	0.869	0.984	1.098	1.213	1.442	1.900	2.36	2.82	3.27	3.73	4.19	4.65
140	0.697	0.742	0.855	0.967	1.080	1.193	1.418	1.868	2.32	2.77	3.22	3.67	4.12	4.57
150	0.686	0.730	0.841	0.951	1.062	1.173	1.395	1.838	2.28	2.72	3.17	3.61	4.05	4.50
175	0.659	0.701	0.807	0.914	1.020	1.127	1.340	1.765	2.19	2.62	3.04	3.47	3.89	4.32
200	0.634	0.675	0.777	0.879	0.982	1.084	1.289	1.698	2.11	2.52	2.93	3.34	3.75	4.16
225	0.610	0.650	0.749	0.847	0.946	1.044	1.242	1.636	2.03	2.43	2.82	3.21	3.61	4.00
250	0.589	0.627	0.722	0.817	0.913	1.008	1.198	1.579	1.959	2.34	2.72	3.10	3.48	3.86
275	0.569	0.606	0.698	0.790	0.881	0.973	1.157	1.525	1.893	2.26	2.63	3.00	3.36	3.73
300	0.550	0.586	0.675	0.764	0.852	0.941	1.119	1.475	1.830	2.19	2.54	2.90	3.25	3.61
350	0.516	0.550	0.633	0.716	0.800	0.883	1.050	1.384	1.717	2.05	2.38	2.72	3.05	3.39
400	0.486	0.518	0.596	0.675	0.753	0.832	0.989	1.303	1.618	1.932	2.25	2.56	2.87	3.19
450	0.459	0.489	0.563	0.638	0.712	0.786	0.934	1.232	1.529	1.826	2.12	2.42	2.72	3.01
500	0.436	0.464	0.534	0.604	0.675	0.745	0.886	1.167	1.449	1.731	2.01	2.29	2.58	2.86
550	0.414	0.441	0.508	0.575	0.641	0.708	0.842	1.110	1.377	1.645	1.912	2.18	2.45	2.72
600	0.394	0.420	0.484	0.547	0.611	0.675	0.802	1.057	1.312	1.567	1.822	2.08	2.33	2.59

Typical properties of gases

Gas or vapour	Molecular weight M	Ratio of specific heats k (14.7 psia)	Coefficient C*	Specific gravity	Critical pressure psia	Critical temperature (°R) (°F+460)
Acetylene	26.04	1.25	342	0.899	890	555
Air	28.97	1.40	356	1.000	547	240
Ammonia	17.03	1.30	347	0.588	1638	730
Argon	39.94	1.66	377	1.379	706	272
Benzene	78.11	1.12	329	2.696	700	1011
N-butane	58.12	1.18	335	2.006	551	766
Iso-butane	58.12	1.19	336	2.006	529	735
Carbon dioxide	44.01	1.29	346	1.519	1072	548
Carbon disulphide	76.13	1.21	338	2.628	1147	994
Carbon monoxide	28.01	1.40	456	0.967	507	240
Chlorine	70.90	1.35	352	2.447	1118	751
Cyclohexane	84.16	1.08	325	2.905	591	997
Ethane	30.07	1.19	336	1.038	708	550
Ethyl alcohol	46.07	1.13	330	1.590	926	925
Ethyl chloride	64.52	1.19	336	2.227	766	829
Ethylene	28.03	1.24	341	0.968	731	509
Freon 11	137.37	1.14	331	4.742	654	848
Freon 12	120.92	1.14	331	4.174	612	694
Freon 22	86.48	1.18	335	2.985	737	665
Freon 114	170.93	1.09	326	5.900	495	754
Helium	4.02	1.66	377	0.139	33	10
N-heptane	100.20	1.05	321	3.459	397	973
Hexane	86.17	1.06	322	2.974	437	914
Hydrochloric acid	36.47	1.41	357	1.259	1198	584
Hydrogen	2.02	1.41	357	0.070	188	60
Hydrogen chloride	36.47	1.41	357	1.259	1205	585
Hydrogen sulphide	34.08	1.32	349	1.176	1306	672
Methane	16.04	1.31	348	0.554	673	344
Methyl alcohol	32.04	1.20	337	1.106	1154	924
Methyl butane	72.15	1.08	325	2.491	490	829
Methyl chloride	50.49	1.20	337	1.743	968	749
Natural gas (typical)	19.00	1.27	344	0.656	671	375
Nitric oxide	30.00	1.40	356	1.036	956	323
Nitrogen	28.02	1.40	356	0.967	493	227
Nitrous oxide	44.02	1.31	348	1.520	1054	557
N-octane	114.22	1.05	321	3.943	362	1025
Oxygen	32.00	1.40	356	1.105	737	279
N-pentane	72.15	1.08	325	2.491	490	846
Iso-pentane	72.15	1.08	325	2.491	490	829
Propane	44.09	1.13	330	1.522	617	666
Sulfur dioxide	64.04	1.27	344	2.211	1141	775
Toluene	92.13	1.09	326	3.180	611	1069

*If 'C' is not known, then use C = 315.

Pipe dimensions

BS 3505 for PVC-U pipe: inch

Diameter					Wall thickness								
					Class C 9.0 bar			Class D 12.0 bar			Class E 15.0 bar		
Nominal size	Mean outside diameter		Individual outside diameter		Average value	Individual value		Average value	Individual value		Average value	Individual value	
	min.	max.	min.	max.	max.	min.	max.	max.	min.	max.	max.	min.	max.
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm
$^3/_8$	17.0	17.3	17.0	17.3	—	—	—	—	—	—	1.9	1.5	1.9
$^1/_2$	21.2	21.5	21.2	21.5	—	—	—	—	—	—	2.1	1.7	2.1
$^3/_4$	26.6	26.9	26.6	26.9	—	—	—	—	—	—	2.5	1.9	2.5
1	33.4	33.7	33.3	33.8	—	—	—	—	—	—	2.7	2.2	2.7
$1^1/_4$	42.1	42.4	42.0	42.5	—	—	—	2.7	2.2	2.7	3.2	2.7	3.2
$1^1/_2$	48.1	48.4	48.0	48.5	—	—	—	3.0	2.5	3.0	3.7	3.1	3.7
2	60.2	60.5	60.0	60.7	3.0	2.5	3.0	3.7	3.1	3.7	4.5	3.9	4.5
3	88.7	89.1	88.4	89.4	4.1	3.5	4.1	5.3	4.6	5.3	6.5	5.7	6.6
4	114.1	114.5	113.7	114.9	5.2	4.5	5.2	6.8	6.0	6.9	8.3	7.3	8.4
5	140.0	140.4	139.4	141.0	6.3	5.5	6.4	8.3	7.3	8.4	10.1	9.0	10.4
6	168.0	168.5	167.4	169.1	7.5	6.6	7.6	9.9	8.8	10.2	12.1	10.8	12.5
8	218.8	219.4	218.0	220.2	8.8	7.8	9.0	11.6	10.3	11.9	14.1	12.6	14.5
10	272.6	273.4	271.6	274.4	10.9	9.7	11.2	14.3	12.8	14.8	17.5	15.7	18.1
12	323.4	324.3	322.2	325.5	12.9	11.5	13.3	17.0	15.2	17.5	20.8	18.7	21.6
14	355.0	356.0	353.7	357.3	14.1	12.6	14.5	18.6	16.7	19.2	22.8	20.5	23.6
16	405.9	406.9	404.3	408.5	16.2	14.5	16.7	21.1	19.0	21.9	26.0	23.4	27.0
18	456.7	457.7	454.9	459.5	18.2	16.3	18.8	23.8	21.4	24.6	—	—	—
20	507.5	508.5	505.4	510.6	20.2	18.1	20.9	—	—	—	—	—	—
24	609.1	610.1	606.5	612.7	24.1	21.7	25.0	—	—	—	—	—	—

BS 5391 for ABS pipe: inch

Diameter					Wall thickness									
Nominal size	Mean outside diameter*		Individual outside diameter		Class B 6.0 bar		Class C 9.0 bar		Class D 15.0 bar		Class E 15.0 bar		Class T† 12.0 bar	
	min.	max.	min.	max.	min.	max.	min.	max.	min.	max.	min.	max.	min.	max.
	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm
3/8	17.0	17.3	17.0	17.3	—	—	—	—	—	—	1.6	1.8	3.4	3.6
1/2	21.2	21.5	21.2	21.5	—	—	—	—	—	—	1.9	2.1	3.5	3.7
3/4	26.6	26.9	26.6	26.9	—	—	—	—	—	—	2.4	2.6	3.5	3.7
1	33.4	33.7	33.4	33.7	—	—	1.9	2.1	2.5	2.7	3.0	3.3	4.2	4.5
1 1/4	42.1	42.4	42.0	42.4	—	—	2.4	2.6	3.1	3.4	3.8	4.1	5.1	5.5
1 1/2	48.1	48.4	48.0	48.5	—	—	2.7	3.0	3.6	3.9	4.4	4.7	5.8	6.2
2	60.2	60.5	60.0	60.7	—	—	3.4	3.7	4.5	4.9	5.4	5.8	7.0	7.4
3	88.7	89.1	88.4	89.4	—	—	5.0	5.3	6.5	6.9	8.0	8.5	—	—
4	114.1	114.5	113.7	114.9	—	—	6.4	6.9	8.4	8.9	10.3	10.9	—	—
6	168.0	168.5	167.4	169.1	6.1	6.4	9.4	10.4	12.3	13.3	—	—	—	—
8	218.8	219.4	218.1	220.2	8.4	8.8	12.2	13.2	—	—	—	—	—	—

*'Mean outside diameter' of a pipe is taken to be the arithmetic mean of any two perpendicularly opposed individual outside diameters. Alternatively, the mean outside diameter mat be determined by means of a circumference tape.

†Class T pipe is intended only for threading. Its maximum sustained working pressure (12 bar) applies when threading is carried out in accordance with BS 21.

PVC-U, PVC-C and ABS metric pipe sizes and tolerances

Size	6 bar	10 bar	16 bar	25 bar
o.d.	wall e mm	wall e mm	wall e mm	wall e mm
6	—	—	1.0	1.0
8	—	—	1.0	1.0
10	—	—	1.0	1.0
12	—	—	1.0	1.0
16	—	—	1.2	1.8
20	—	—	1.5	2.3
25	—	1.5	1.9	2.8
32	—	1.8	2.4	—
40	1.8	1.9	3.0	—
50	1.8	2.4	3.7	—
63	1.9	3.0	4.7	—
75	2.2	3.6	5.6	—
90	2.7	4.3	6.7	—
110	3.2	5.3	8.2	—
125	3.7	6.0	9.3	—
140	4.1	6.7	10.4	—
160	4.7	7.7	11.9	—
180	5.3	8.6	13.4	—
200	5.9	9.6	14.9	—
225	6.6	10.8	16.7	—
250	7.3	11.9	18.6	—
280	8.2	13.4	20.8	—
315	9.2	15.0	23.4	—
355	10.4	16.9	26.3	—
400	11.7	19.1	29.7	—

Tolerances on wall thickness	
o.d.	±
1.0	0.3
1.2–2.0	0.4
2.2–3.0	0.5
3.2–4.0	0.6
4.1–5.0	0.7
5.3–6.0	0.8
6.2–7.0	0.9
7.2–8.0	1.0
8.2–9.0	1.1
9.2–10.0	1.2
10.4–11.0	1.3
11.7–11.9	1.4
12.3–12.4	1.5
13.2–14.0	1.6
14.6–15.0	1.7
15.6–15.7	1.8
16.4–16.9	1.9
17.7–17.8	2.0
18.4–18.6	2.1
19.1–20.0	2.2
20.7–20.8	2.3
21.5	2.4
22.3	2.5
23.3–23.9	2.6
25	2.7
26.3 and 26.7	2.9
27.8	3.0
29.5–30.0	3.2

Outside diameter tolerances	
o.d.	±
5–63	0.2
75–125	0.3
140–200	0.4
225–250	0.5
280–315	0.6
355–400	0.7

PP-H and PE metric pipe sizes and tolerances

Size	2.5 bar	6 bar	10 bar
o.d.	wall e mm	wall e mm	wall e mm
16	—	—	2.0
20	—	1.8	2.5
25	—	1.8	2.7
32	—	2.0	3.0
40	—	2.3	3.7
50	1.8	2.9	4.6
63	1.8	3.6	5.8
75	1.9	4.3	6.9
90	2.2	5.1	8.2
110	2.7	6.3	10.0
125	3.1	7.1	11.4
140	3.5	8.0	12.8
160	3.9	9.1	14.6
180	4.4	10.2	16.4
200	4.9	11.4	18.2
225	5.5	12.8	20.5
250	6.1	14.2	22.8
280	6.9	15.9	25.5
315	7.7	17.9	28.7
355	8.7	20.1	32.3
400	9.8	22.7	36.4

Tolerances on wall thickness	
o.d.	±
1.8–2.0	0.4
2.2–3.0	0.5
3.1–3.9	0.6
4.3–4.9	0.7
5.1–5.8	0.8
6.1–7.0	0.9
7.1–8.0	1.0
8.2–8.7	1.1
9.1–10.0	1.2
10.2–11.0	1.3
11.4	1.4
12.2 and 12.8	1.5
13.7	1.6
14.2 and 14.6	1.7
15.4 and 15.9	1.8
16.4	1.9
17.4 and 17.9	2.0
18.2	2.1
19.3 and 19.6	2.2
20.1 and 20.5	2.3
21.6	2.4
22.0 and 22.8	2.5
24.3 and 24.4	2.7
25.5	2.8
27.4	3.0
28.3 and 28.7	3.1
30.8	3.3
31.7	3.4
32.3	3.5
34.7	3.7
35.7	3.8
36.4	3.9

Outside diameter tolerances	
o.d.	±
10–32	0.3
40	0.4
50	0.5
63	0.6
75	0.7
90	0.9
110	1.0
125	1.2
140	1.3
160	1.5
180	1.7
200	1.8
225	2.1
250	2.3
280	2.6
315	2.9
355	3.2
400	3.6

Support spacing (ft) for PVC pipe*

Horizontal pipe-systems support spacings are greatly influenced by operating temperature. The charts show the recommended support spacing according to size, schedule and operating temperatures. Do not clamp supports tightly — this restricts axial movement of the pipe. If short spacing is necessary, continuous supports may be more economical. Charts are based on liquids up to 1.00 specific gravity, but do not include concentrated loads, nor do they include allowance for aggressive reagents.

Pipe size	SCHEDULE 40 temperature (°F)					SCHEDULE 80 temperature (°F)					SCHEDULE 120 temperature (°F)				
in	60	80	100	120	140	60	80	100	120	140	60	80	100	120	140
¼	4	3½	3½	2	2	4	4	3½	2½	2					
⅜	4	4	3½	2½	2	4½	4½	4	2½	2½					
½	4½	4½	4	2½	2½	5	4½	4	3	2½	5	5	4½	3	2½
¾	5	4½	4	2½	2½	5½	5	4½	3	2½	5½	5	4½	3	3
1	5½	5	4½	3	2½	6	5½	5	3½	3	6	5½	5	3½	3
1¼	5½	5½	5	3	3	6	6	5½	3½	3	6½	6	5½	3½	3½
1½	6	5½	5	3½	3	6½	6	5½	3½	3½	6½	6½	6	4	3½
2	6	5½	5	3½	3	7	6½	6	4	3½	7½	7	6½	4	3½
2½	7	6½	6	4	3½	7½	7½	6½	4½	4	8	7½	7	4½	4
3	7	7	6	4	3½	8	7½	7	4½	4	8½	8	7½	5	4½
3½	7½	7	6½	4	4	8½	8	7½	5	4½	9	8½	7½	5	4½
4	7½	7	6½	4½	4	9	8½	7½	5	4½	9½	9	8½	5½	5
5	8	7½	7	4½	4	9½	9	8	5½	5	10½	10	9	6	5½
6	8½	8	7½	5	4½	10	9½	9	6	5	11½	10½	9½	6½	6
8	9	8½	8	5	4½	11	10½	9½	6½	5½					
10	10	9	8½	5½	5	12	11	10	7	6					
12	11½	10½	9½	6½	5½	13	12	10½	7½	6½					
14	12	11	10	7	6	13½	13	11	8	7					
16	12½	11½	10½	7½	6½	14	13½	11½	8½	7½					
			SDR 41					SDR 26							
18	13	12	11	8	7	14½	14	12	9	8					
20	13½	12½	11½	8½	7½	15	14½	12½	9½	8½					
22	14	13	12	9	8	15½	15	13	10	9					

*Although support spacing is shown at 140°F, consideration should be given to the use of CPVC or continuous support above 120°F. The possibility of temperature overrides beyond regular working temperatures and cost may make either of the alternatives more desirable.
This chart is based on continuous spans and for iniusulated line carrying fluids of specific gravity up to 1.00.

Support spacing (ft) for CPVC pipe*

Pipe size	SCHEDULE 40 temperature °F						SCHEDULE 80 temperature °F					
in	73	100	120	140	160	180	73	100	120	140	160	180
1/2	5	4 1/2	4 1/2	4	2 1/2	2 1/2	5 1/2	5	4 1/2	4 1/2	3	2 1/2
3/4	5	5	4 1/2	4	2 1/2	2 1/2	5 1/2	5 1/2	5	4 1/2	3	2 1/2
1	5 1/2	5 1/2	5	4 1/2	3	2 1/2	6	6	5 1/2	5	3 1/2	3
1 1/4	5 1/2	5 1/2	5 1/2	5	3	3	6 1/2	6	6	5 1/2	3 1/2	3
1 1/2	6	6	5 1/2	5	3 1/2	3	7	6 1/2	6	5 1/2	3 1/2	3 1/2
2	6	6	5 1/2	5	3 1/2	3	7	7	6 1/2	6	4	3 1/2
2 1/2	7	7	6 1/2	6	4	3 1/2	8	7 1/2	7 1/2	6 1/2	4 1/2	4
3	7	7	7	6	4	3 1/2	8	8	7 1/2	7	4 1/2	4
3 1/2	7 1/2	7 1/2	7	6 1/2	4	4	8 1/2	8 1/2	8	7 1/2	5	4 1/2
4	7 1/2	7 1/2	7	6 1/2	4 1/2	4	8 1/2	9	8 1/2	7 1/2	5	4 1/2
6	8 1/2	8	7 1/2	7	5	4 1/2	10	9 1/2	9	8	5 1/2	5
8	9 1/2	9	8 1/2	7 1/2	5 1/2	5	11	10 1/2	10	9	6	5 1/2
10	10 1/2	10	9 1/2	8	6	5 1/2	11 1/2	11	10 1/2	9 1/2	6 1/2	6
12	11 1/2	10 1/2	10	8 1/2	6 1/2	6	12 1/2	12	11 1/2	10 1/2	7 1/2	6 1/2

This chart is based on continuous spans and for ininsulated line carrying fluids of specific gravity up to 1.00.

*The data furnished herein is based on information furnished by manufacturers of the raw material. This information may be considered as a basis for recommendation, but not as a guarantee. Materials should be tested under actual service to determine duitability for a particular purpose.

Pipe-bracket spacing for liquids with a density of ≤ 1 g/cm³ and for gases

PVC-U

d		Pipe bracket intervals L in cm at				
mm	in	20°C	30°C	40°C	50°C	60°C
16	3/8	80	70	50	Continuous support	
20	1/2	90	80	60		
25	3/4	95	85	65	55	40
32	1	105	90	70	60	45
40	1 1/4	120	110	90	70	55
50	1 1/2	140	130	110	85	65
63	2	150	140	120	95	70
75	2 1/2	165	155	135	110	80
90	3	180	170	150	125	95
110	4	200	190	170	145	115
125	—	210	200	185	160	125
140	5	225	215	195	170	140
160	6	240	230	210	185	155
200	7	255	240	220	200	170
225	8	270	260	240	215	185

PVC-C

d		Pipe bracket intervals L in cm at						
mm	in	20°C	30°C	40°C	50°C	60°C	70°C	80°C
16	3/8	100	95	90	85	75	67	60
20	1/2	115	110	100	95	87	77	70
25	3/4	120	115	110	100	90	80	70
32	1	135	125	120	110	100	90	80
40	1 1/4	150	140	130	125	115	105	90
50	1 1/2	165	160	150	140	130	120	110
63	2	185	175	165	160	150	135	125
75	2 1/2	205	195	185	175	165	150	135
90	3	225	210	200	190	180	165	150
110	4	250	235	220	210	195	180	165
160	6	300	285	270	255	240	220	200
225	8	355	335	320	300	280	260	235

ABS

d		Pipe bracket intervals L in cm at				
mm	in	20°C	30°C	40°C	50°C	60°C
16	3/8	70	65	60	55	45
20	1/2	80	70	65	60	50
25	3/4	85	80	75	65	60
32	1	100	90	85	75	65
40	1 1/4	110	100	95	85	75
50	1 1/2	115	110	100	90	80
63	2	130	120	110	100	85
90	3	160	145	135	120	105
110	4	180	165	155	135	120
160	6	230	210	200	175	155

The bracket spacings shown in the table are for Class C and PN 10 pipe and are given in cm.

For other pipe classes the figures must be multiplied by the following factors:

Class B 0.90
Class C 1.05
Class E 1.09

PP

d	Pipe bracket intervals L in cm at						
mm	20°C	30°C	40°C	50°C	60°C	80°C	100°C
16	75	70	70	65	65	55	40
20	80	75	70	70	65	60	45
25	85	85	85	80	75	70	50
32	100	95	95	90	85	75	55
40	110	110	105	100	95	85	60
50	125	120	115	110	105	90	70
63	140	135	130	125	120	105	80
75	155	150	145	135	130	115	85
90	165	165	155	150	145	125	95
110	185	180	175	165	160	140	105
125	200	190	185	180	170	150	110
140	210	205	195	190	180	155	115
160	225	225	210	200	190	165	125
180	240	240	225	215	200	170	130
200	250	250	235	225	215	185	135
225	265	260	250	240	230	200	145
250	280	275	265	255	240	210	200
315	315	305	295	285	270	235	225

PB

d	Pipe bracket intervals L in cm at						
mm	20°C	30°C	40°C	50°C	60°C	80°C	100°C
16	70	69	68	66	64	60	55
20	78	77	76	74	72	68	61
25	81	80	79	77	75	71	64
32	93	91	90	88	85	80	73
40	103	102	100	98	95	90	81
50	115	114	112	109	106	100	90
63	130	128	126	123	119	112	102
75	141	139	137	133	130	122	110
90	154	152	150	146	142	134	121
110	188	186	184	179	173	164	148

PE

d	Pipe bracket intervals L in cm at				
mm	20°C	30°C	40°C	50°C	60°C
20	75	70	65	65	60
25	80	80	75	70	65
32	90	90	85	80	75
40	100	100	95	90	85
50	115	110	105	100	95
63	130	125	120	115	105
75	140	135	130	125	115
90	155	150	145	135	130
110	170	165	160	150	140
125	185	175	170	160	150
140	195	185	180	170	155
160	210	200	190	180	170
200	235	220	210	200	186
225	250	235	220	210	200

PVDF

d	Pipe bracket intervals L in cm at						
mm	20°C	40°C	60°C	80°C	100°C	120°C	140°C
16	75	70	70	65	65	55	40
20	80	75	70	70	65	60	45
25	85	85	85	80	75	70	50
32	100	95	95	90	85	75	55
40	110	110	105	100	95	85	60
50	125	120	115	110	105	90	70
63	140	135	130	125	120	105	80
75	155	150	145	135	130	115	85
90	165	165	155	150	145	125	95
110	185	180	175	165	160	140	105
125	200	190	185	180	170	150	110
140	210	205	195	190	180	155	115
160	225	225	210	200	190	165	125
200	250	250	235	225	215	185	135
225	265	260	250	240	230	200	145

SECTION 10

Author's Acknowledgements

Author's Acknowledgments

Abacus Valves Mfg Ltd
ASCO / Joucomatic
N H Bennett
Biwater Industries
British Standards Institute
Buracco SA
BVAMA
Crosby Valve Inc
CUES
Danfoss
Dow Chemical Company
Dresser Industries
Durabla Fluid Technology Inc.
Fisher–Rosemount
FMC Corporation
George Fischer
Griffin Pipe Products
Harvel Plastics Inc.
Hindle Cockburns Ltd
Hitachi Valves Ltd
IMI Bailey Birkett Ltd
KSB
Microfinish Valves Limited
Neles-Jamesbury
OMB SpA
Pipes & Pipelines International
Realm Products Ltd
Rotork Controls Ltd
A Searle
Shafer Products
Spirax–Sarco
P. Stockford
Vanessa Srl
Victualic International
T D Williamson Inc

SECTION 11

Buyer's Guide to Valves and Pipes

Classified Index by Product Category

Valve & Pipe Equipment

Valves & Actuators

Cast Steel Valves
PCC Flow Technologies
NAF AB
Posi-Flate
Hattersley Newman Hender Ltd

Check Valves
PCC Flow Technologies
Changdel Industrial Co. Ltd
ASCO/Joucomatic (ASCO Controls BV)
Hindle Cockburns Ltd
NAF AB
Spirax-Sarco Limited
Victaulic Company
Hattersley Newman Hender Ltd
Georg Fischer AG

Control Valves
PCC Flow Technologies
NAF AB
Posi-Flate
Spirax-Sarco Limited
Victaulic Company
Hattersley Newman Hender Ltd
Georg Fischer AG

Diaphragm Valves
ASCO/Joucomatic (ASCO Controls BV)
Hattersley Newman Hender Ltd
Georg Fischer AG

Diverter Valves
Hindle Cockburns Ltd

Electronically Operated Valves
NAF AB

Fire Safe Valves
NAF AB

Float Control Valves
Victaulic Company
Hattersley Newman Hender Ltd

Flow Valves
Posi-Flate
Victaulic Company
Hattersley Newman Hender Ltd

Foot Valves
Hattersley Newman Hender Ltd

Gas Valves
ASCO/Joucomatic (ASCO Controls BV)
Hindle Cockburns Ltd
Hattersley Newman Hender Ltd
Georg Fischer AG

Gate Valves
Hindle Cockburns Ltd
NAF AB
Hattersley Newman Hender Ltd

Gear Operated Valves
PCC Flow Technologies
NAF AB
Posi-Flate
Victaulic Company
Hattersley Newman Hender Ltd

Globe Valves
Hindle Cockburns Ltd
NAF AB
Hattersley Newman Hender Ltd

High Pressure Valves
PCC Flow Technologies
Hindle Cockburns Ltd
ASCO/Joucomatic (ASCO Controls BV)
Hattersley Newman Hender Ltd

High Temperature Valves
PCC Flow Technologies
Hindle Cockburns Ltd
ASCO/Joucomatic (ASCO Controls BV)
NAF AB
Posi-Flate

Hydraulically Operated Valves
Hindle Cockburns Ltd

Intrinsically Safe Valves
ASCO/Joucomatic (ASCO Controls BV)

Isolating Valves
ASCO/Joucomatic (ASCO Controls BV)
NAF AB
Spirax-Sarco Limited
Hattersley Newman Hender Ltd

Level Control Valves
Hattersley Newman Hender Ltd

Manifold Valves
ASCO/Joucomatic (ASCO Controls BV)

Metal Valves
Hindle Cockburns Ltd
NAF AB

Metering Valves
Hindle Cockburns Ltd
Hattersley Newman Hender Ltd

Mixing Valves
Spirax-Sarco Limited
Georg Fischer AG

Swingcheck Valves
PCC Flow Technologies
NAF AB
Hattersley Newman Hender Ltd

Tank Valves
ASCO/Joucomatic (ASCO Controls BV)

Temperature Control Valves
NAF AB
Spirax-Sarco Limited
Hattersley Newman Hender Ltd

Throttling Valves
NAF AB
Hattersley Newman Hender Ltd

Triple Offset Valves
PCC Flow Technologies
NAF AB

Vacuum Valves
ASCO/Joucomatic (ASCO Controls BV)

Knife Gate Valves
PCC Flow Technologies

Pipes & Piping

ABS Pipe
Georg Fischer AG

CPVC Pipe
Georg Fischer AG

Dual/Double Containment Pipe
Georg Fischer AG

Flexible Pipe
Georg Fischer AG

PE Pipe
Georg Fischer AG

Pipe Clamps
Georg Fischer AG

Pipelines Couplings
Victaulic Company
Georg Fischer AG

Pipe Cutting Equipment
Georg Fischer AG

Pipe Fittings — Iron and Steel
Victaulic Company
Georg Fischer AG

Pipe Fittings—Plastics
Victaulic Company
Georg Fischer AG

Pipe Fusion Equipment
Georg Fischer AG

Pipe Joints
Georg Fischer AG

Plastic Pipe
Victaulic Company
Georg Fischer AG

Pressfit Pipe
Victaulic Company

PVC—C Pipe
Georg Fischer AG

PVC—U Pipe
Georg Fischer AG

Thermoplastic Pipe
Georg Fischer AG

Ancillary Equipment & Services

Insulated Valves Covers
Hattersley Newman Hender Ltd

Leak Detection Equipment
Spirax-Sarco Limited

Noise Control
NAF AB

Packings
Latty International SA

Positioners
NAF AB
Spirax-Sarco Limited

Process Controllers
NAF AB

Regulators
NAF AB
Hattersley Newman Hender Ltd

Seals
Latty International SA

Sealing Materials
Latty International SA

Silencers
NAF AB

Steam Traps
Spirax-Sarco Limited
Hattersley Newman Hender Ltd

Traps/Drainers
Hattersley Newman Hender Ltd

Valve Position Indicators
NAF AB
Hattersley Newman Hender Ltd

Valve Testing
Hindle Cockburns Ltd
Hattersley Newman Hender Ltd

Industries & Applications

Water

Boiler Feed
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)
NAF AB
Spirax-Sarco Limited
Hattersley Newman Hender Ltd

Brackish Water
PCC Flow Technologies

Cooling Water
PCC Flow Technologies
Hattersley Newman Hender Ltd
ASCO/Joucomatic (ASCO Controls BV)
NAF AB

Drinking Water
PCC Flow Technologies
Hattersley Newman Hender Ltd
ASCO/Joucomatic (ASCO Controls BV)

Sea Water
PCC Flow Technologies
Hattersley Newman Hender Ltd

Sewage
PCC Flow Technologies
Hattersley Newman Hender Ltd
ASCO/Joucomatic (ASCO Controls BV)

Waste Water
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)
Hattersley Newman Hender Ltd

Food & Beverage

Beverage
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)
Hindle Cockburns Ltd
Posi-Flate
Hattersley Newman Hender Ltd

Brewery
PCC Flow Technologies
Posi-Flate

Dairy
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)

Distillers
PCC Flow Technologies
Hattersley Newman Hender Ltd

General Food
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)
NAF AB
Posi-Flate

Sugar
PCC Flow Technologies
NAF AB
Hattersley Newman Hender Ltd

Wine
PCC Flow Technologies
Hattersley Newman Hender Ltd

Chemical & Process

Catalysts
PCC Flow Technologies
Hindle Cockburns Ltd
Posi-Flate

Chlor Alkali
PCC Flow Technologies
Hindle Cockburns Ltd
NAF AB

Clarifying
PCC Flow Technologies
Hindle Cockburns Ltd

Emulsions
PCC Flow Technologies
Hindle Cockburns Ltd

Fuel Oil (Heavy)
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)
Hindle Cockburns Ltd

Fuel Oil (Light)
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)
Hindle Cockburns Ltd

Grease/Lubricating Oil
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)
Hindle Cockburns Ltd
Hattersley Newman Hender Ltd

Industrial Chemicals
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)
Hindle Cockburns Ltd
NAF AB
Posi-Flate

Inks & Dyes
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)
Hindle Cockburns Ltd
Posi-Flate

Oils
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)
Hindle Cockburns Ltd

Paints
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)
Hindle Cockburns Ltd
Posi-Flate

Polymers
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)
Hindle Cockburns Ltd
Posi-Flate

Process
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)
Hindle Cockburns Ltd
Posi-Flate

Resins
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)
Hindle Cockburns Ltd
Posi-Flate

Solvents & Bleaching
PCC Flow Technologies
Hindle Cockburns Ltd
Posi-Flate

Pulp & Paper
NAF AB

Gas & Pollution

Cryogenic Gas
PCC Flow Technologies
ASCO/Joucomatic (ASCO Controls BV)
Hindle Cockburns Ltd

Dioxins
PCC Flow Technologies
Hindle Cockburns Ltd

Effluent
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AS3–AS50—Electric part-turn actuators—Auma Werner Riester GmbH & Co KG

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DREHME—Electric actuators for remote valve operation- EMG Elektro-Mechanik GmbH

EMG—Electric actuators for remote valve operation—EMG Elektro-Mechanik GmbH

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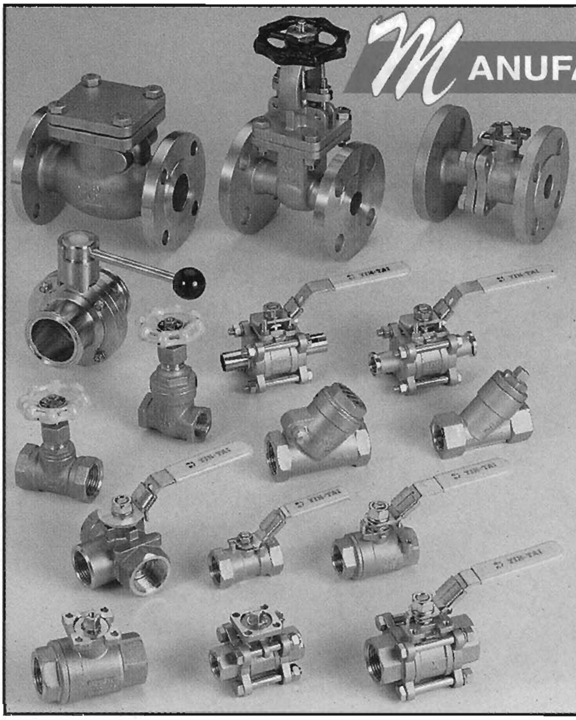

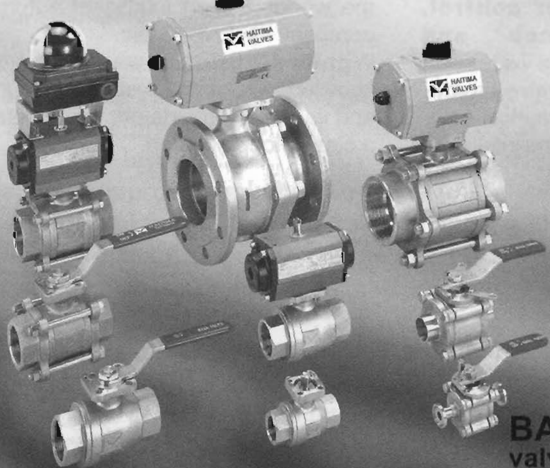

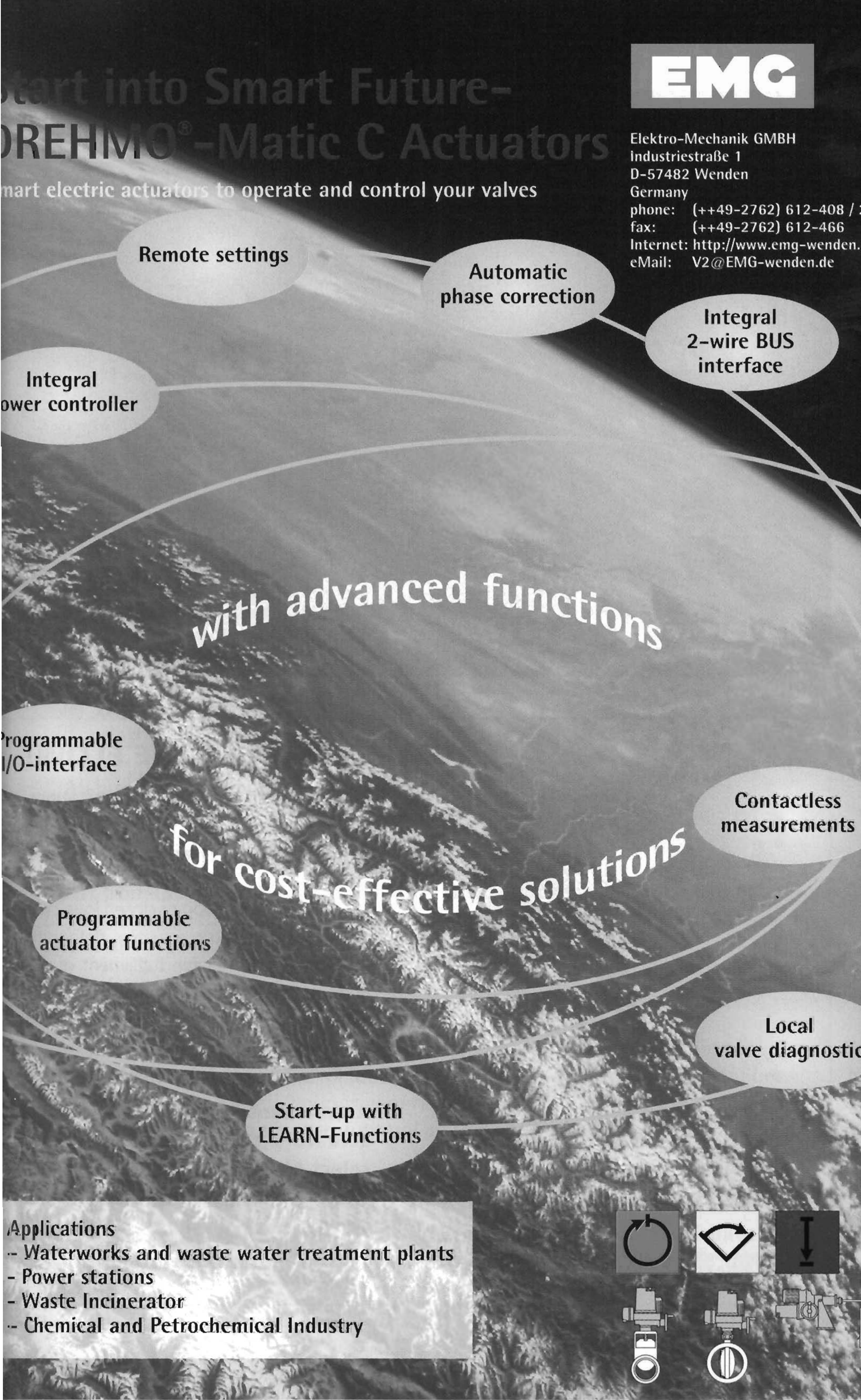
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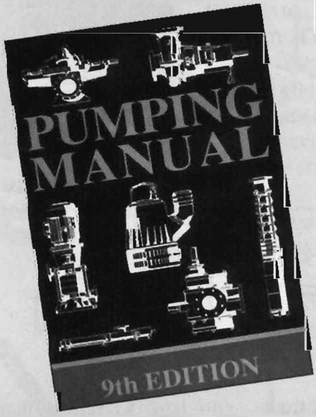

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